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A Series of Books in Psychology

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Validity of Measurement

To set about measuring a psychological characteristic of individuals, we first define it and then develop operations designed to yield quantitative descriptions of the extent to which individuals possess or manifest it. Since the definition is only a blueprint, which may or may not be exactly followed in developing the operations of measurement, we may well wonder about the degree to which the properties measured by a set of operations *in fact* correspond to the characteristic as defined. Thus before a set of operations is accepted, it may be necessary to obtain some indications that it measures what is intended.

Also we are often interested in predicting or forecasting behavior. For example, in connection with counseling high school students, we may wish to predict from their scholastic aptitude test score the level of academic success they are likely to attain if they go to college. Similarly, in appraising applicants for a sales job, we might wish to forecast their probable success from the scores they earn on a test of sociability. In these circumstances we have some clearly specified type of behavior or characteristic, such as college grades earned or dollar value of merchandise sold, measured by one set of operations, and we are concerned about the accuracy with which we can predict it from the traits measured by another set of operations.

It is from these and similar situations that the problem termed *validity of measurement* arises. In general it is apparent that validity pertains to

the nature of the variable being measured by a set of operations. But it will also be apparent that there is not a high degree of agreement on what constitutes the problem of validity. Consequently there also is no consensus on what constitutes proper *validation* (that is, the procedure by means of which the validity of tests or other operations of measurement is determined).

Defining Validity and Validation

As is the case with the reliability of measurement, the term validity of measurement means different things to different people. While it is possible to formulate different definitions of reliability in reasonably precise ways, this is not always the case with the validity of measurement.

Recall the variety of purposes, both theoretical and practical, for which individuals are measured. Each of these many purposes has led to somewhat different notions about what the validity of measurement is and to the development of different procedures for determining validity. Keeping these ambiguities in mind, we would like to present the following framework for thinking about validity problems. It is meant to be reasonably orthodox.

Historically the most common definition of validity is that it refers to the extent to which a test or a set of operations measures what it is supposed to measure. A more recent, and perhaps better, definition is the "official" one given in the American Psychological Association, American Educational Research Association, National Council on Standards for Educational and Psychological Tests (1974): "[V]alidity refers to the appropriateness of inferences from test scores or other forms of assessment" (p. 25). Loosely paraphrased, this means: Given a set of specific questions we want a psychological measure to help answer, how useful or appropriate (that is, valid) are the answers (that is, the information) provided by the test scores? One advantage of this definition is that it ties the degree of validity of a measure directly to the extent to which it is appropriate for answering specific questions. For example, how valid is a measure for predicting (that is, inferring) future job success? How valid is a measure for assessing an individual's degree of dominance? How valid is a measure for assessing how much an individual has learned in a course? There are literally dozens of such questions that can be asked, and the methods used to validate the appropriateness of a measure for answering them would most likely also differ. It follows from this definition that it is difficult to talk about different "types" of validation or validity evidence. The nature of validity varies to greater or lesser degree as a function of the specific

question being considered. Nevertheless talking about types of validity is what people in the field have traditionally done and we will follow suit.

Prior to 1954 (seems like a long time ago) the concept of validity was in considerable disarray. There were almost as many definitions and varieties of validity as there were people interested in psychometric theory. Something had to be done. Naturally a committee was formed (in this case by the American Psychological Association) to look into the matter, and in 1954 the APA's *Technical Recommendations for Psychological Tests and Diagnostic Techniques* were published. This document has gone through two major revisions (APA, 1966; APA, AERA, NCME, 1974) and work is beginning on a third (as of this writing).

Since the *Recommendations* reflect the consensus of a great many of the people who work in this field, we would like to briefly discuss the major types of validity and/or validation as they are portrayed in this document. However, the reader should keep in mind that, as we argued above, any such classification of validity types is potentially misleading. The nature of validity, and the data it demands for its substantiation, may vary with the nature of the specific questions being asked. Thinking of validity in terms of "types" may mislead an investigator as to how he or she should try to establish the validity of a measure. Nevertheless discussing validity types does help get the conversation started.

Criterion-Related Validity

In many instances the hypotheses to be tested or the questions to be asked about the validity of a test or other measure concern how well the scores on the test correspond to, are related to, or predict scores on another variable called a *criterion*. The criterion is the individual characteristic (variable) of real interest. In such instances, the test is of interest only insofar as it is related to the criterion, and can appropriately be used as a substitute for it. In general, we will refer to this substitute variable as the *predictor*, although sometimes the word is a bit of a misnomer. If it were always possible to obtain criterion scores, there would be no need for the predictor measure and no need for criterion-related validities. However, it is unfortunately the case that criterion scores are not always obtainable or available. They may already have occurred in the past and thus are no longer available, they may not occur until sometime in the future, or they may simply be too expensive or too dangerous to obtain for each person under consideration.

In U.S. society some examples of criteria that have been deemed

important enough to worry about in terms of how strongly they are related to other, more readily available measures are the following. Certainly one of the most important is job performance. How well will someone do if he or she is hired? How well will *you* do if you take a particular job? Notice the distinction here between the importance of job performance from the perspective of the organization doing the hiring and the perspective of the individual seeking the job. Performance may be defined and measured differently by each. The organization may be interested in how much you contribute to its profits. You may be interested in whether you will be able to use your skills in such a way that you will feel a strong sense of accomplishment. It follows that different kinds of information may be used to predict each of these two kinds of performance criteria and that their validities would be determined differently. Whether or not a criterion is an appropriate one is first and foremost a value judgment. To be sure, we may worry about the reliability of the criterion measure (for example, the stability of performance ratings over time), but the determination of its actual content is a value judgment, and it should be confronted as such.

In education the analogy to job performance is school or university achievement. When someone enters a college or university, how well will he or she do on various measures of scholastic achievement (for example, grade-point average)?

For both employers and schools a criterion of great importance is how long the individual will stay in the organization or whether he or she will finish the prescribed curriculum. Thus tenure, turnover, and finish/not finish are all examples of criterion measures that reflect individual tendencies to stay in the organization.

The above are all examples of criteria that will occur sometime in the future and thus are currently unavailable. But suppose a decision must be made about the individual *now*. Thus comes the question: Are there economical measures that are available now and that can be used as a substitute?

An example of a currently available criterion that may simply be too expensive to obtain is a full assessment of the degree to which an individual exhibits some form of mental illness. A complete clinical and psychiatric workup is expensive and may take several days. If the task is to screen a large number of people for such symptoms, a cheaper substitute may have to be found. The question then arises as to how strongly the cheaper substitute (for example, a paper-and-pencil personality inventory) is related to the results of the full examination, which is the variable of real interest (that is, the criterion).

Another reason for the unavailability of the criterion is that it may reflect events that occurred in the past and cannot be currently assessed. For example, physicians may attempt to infer whether or not an

individual has had a heart attack on the basis of an electrocardiogram (EKG) record. The EKG (the predictor) is of no real interest in and of itself. It is only of interest in terms of how well it allows the physician to infer whether a particular individual did or did not have a heart attack. To cite another example, it is the task of the jury in criminal cases to infer whether or not an individual committed a crime in the past (the criterion) from the current presentation of the evidence in the courtroom (the predictor).

The criterion-related validity of a test is described in an objective and quantitative fashion by the degree of relationship between predictor scores and criterion scores. Thus the Pearsonian correlation coefficient or its variants, such as the point biserial, biserial, phi, or tetrachoric coefficient, are commonly used to describe the degree of predictive validity. When these coefficients are used to indicate the degree of relationship between predictor and criterion scores, they are termed *validity coefficients*.

It is obvious that the relationship between two variables can be indicated by means other than the coefficient of correlation. A scatter diagram or bivariate distribution graphically depicts the relationship between two variables. The difference between the means of the *Y* (criterion) scores of those individuals earning high and low *X* (predictor) scores also tells us something about the relationship between the two variables. But the coefficient of correlation, the validity coefficient, is most commonly used because it seemingly gives the most precise description of the degree of relationship in a simple and convenient manner. We will come back to this seductive simplicity later.

Again the moral of this entire story is that the criterion is the variable of real interest, and its importance stems from the value judgments of some significant segment of society (for example, the Congress, your boss). The appropriateness with which other variables can be used as a substitute for the criterion speaks to the criterion-related validities of these other variables.

Before discussing different varieties of criterion-related validities, one further point should be made. A particular variable may serve as a criterion on one occasion and as a predictor on another. That is, what's a criterion today might not be tomorrow. For example, a relevant validity problem might be to predict subsequent college grade-point average (GPA) from a scholastic aptitude test taken while an individual is a high school senior. The criterion is college GPA. Four years later an employer may try to predict future performance on job *X* from college GPA. Now performance on job *X* is the criterion, and GPA is the substitute variable (predictor). Still later the employer may want to predict future performance on job *Y* from an assessment of current performance on job *X*. Now performance on job *Y* is the criterion, and

performance on job *X* is the predictor. In sum, which variable is labeled as the criterion is a function of the specific intent of the decisionmaker at a specific time.

Predictive Validity

Predictive validity describes the accuracy with which we can estimate the extent to which some individual characteristic will be manifested in the future from the extent to which the individual currently manifests or possesses some other property. For example, when a high school student is counseled, it is helpful to be able to predict what his or her performance in college is likely to be. Consequently we would be interested in knowing the degree of validity of high school grades for predicting college grades. When applicants for a job are considered, it is helpful to estimate which applicants are most likely to perform well later on. Therefore we would like to know the validity with which scores on a test or other measure administered at the time of hiring predict later performance on the job.

Evidence for predictive validity It is one thing to define predictive validity as the degree to which a current measure (the predictor) is related to (correlated with) the variable of real interest (the criterion), which is not observed until sometime in the future. Collecting the appropriate data so as to compute the required correlation, now termed a *predictive validity coefficient*, is another matter. Spouting definitions is easy; collecting good data is difficult.

Simply put, a predictive validity study involves (1) obtaining an appropriate sample of people, (2) measuring them on the predictor, (3) waiting for the necessary weeks, months, or years to pass, (4) obtaining the criterion scores for the same sample of people, and (5) computing the correlation between the predictor score and the criterion score. The higher the correlation, the greater the predictive validity, *other things being equal*. We shall get to some of these "other things" in a moment.

An example of such a study would be (1) measuring a representative sample of job applicants on the predictor measure (for example, aptitude test or interviewer ratings), (2) hiring everyone, (3) allowing everybody to work on the job for a period that is long enough to go beyond the initial learning phase and thus allowing differences in job proficiency to emerge, (4) assessing job performance with a standardized criterion measure, and (5) computing the correlation between the predictor measure and the job-performance measure. From this point on, there is one important fact that should be remembered. Establishing any kind of criterion-related validity requires that *for at least one sam-*

ple of people both the predictor and the criterion score must be obtained, no matter how difficult or expensive it is. Within the limits of sampling error, we can then infer what the accuracy of the predictor will be in subsequent samples, and use it accordingly. But without data from at least one representative sample, we cannot generate any kind of criterion-related validity.

Obviously there are a number of factors that enhance or detract from the usefulness of a predictive validity study. One of the most important is the *attrition rate*, or the frequency with which people drop out of the validation sample between the time the predictor scores are obtained and the time the criterion scores are obtained. Besides simply decreasing the sample size, and thereby increasing the sampling error inherent in the estimate of the predictive validity coefficient, the attrition may be on a selective and not a random basis. That is, the people who quit the job, drop out of school, and so on, may not be at all representative of the original applicant pool, and thus the people who are left in the sample may not be representative either. As a result, the validity coefficient computed on the sample suffering from selective attrition may be much different from a coefficient computed on the entire population of interest.

Another potential difficulty with predictive validity studies is that it may be hard to wait too long. Many such studies have gone down the drain because the pressures (economic and/or political) to start using the predictor information before it's time are simply too great.

Finally, the nature of the predictive study tends to make it very expensive. Not all organizations are willing to stand the cost of such research.

Concurrent Validity

Sometimes we are interested in the precision with which we can estimate the extent to which the individual *currently* possesses a given trait or behaves in a particular way (the criterion) from the extent to which he or she *currently* possesses some other trait or behaves in some other way (the predictor). We use scores on one variable, then, to estimate scores on another, both variables measuring *present* properties of the individual. It may be too difficult, expensive, or time-consuming to measure the criterion directly, and therefore we prefer to use some other way to obtain quantitative descriptions of that property. This type of validity is often called *concurrent validity*.

If we were interested in measuring proficiency in repairing automobiles, it would be necessary to measure the speed and accuracy with which the individual could rectify all the various automotive dysfunc-

tions, from the fuel system to the differential. We would have to gather a large number of cars manifesting all the sundry mechanical disorders. Each person would be required to do all the repair tasks, and we would have to be sure that each person was presented with exactly the same series of problems. Since it probably would take a number of days to measure the speed and accuracy of a single individual's performance, it would be expensive to do it very often. Consequently, if we found that scores on a paper-and-pencil test of knowledge of automotive mechanics and repair were substantially related to performance scores, we could use them as substitutes for the cumbersome and costly performance measures.

In any event, the touchstone of concurrent validity is that both predictor and criterion are measured at about the same time.

Substitute for predictive validity We noted previously that one drawback to determining predictive validity is that the people footing the bill may not have the patience to wait for the necessary time to pass. Then why not use a concurrent design to estimate validity? For example, if the problem is to estimate the predictive validity of a job-simulation test, why not give the test to a sample of people who are already on the job and measure their job performance at the same time? Such a study might take only a few weeks, while a predictive study might take several years. The analogy in an educational setting would be to give a scholastic aptitude test to a sample of students who had just finished one or more years of college and obtain their grade-point average at the same time. The correlation between these two sets of scores would then be the estimate of how well we could predict future college success when the scholastic aptitude test is given to high school students.

There is nothing inherently wrong in this procedure. The effects of attrition are about the same since the individuals who drop out of a predictive validity sample are the same people who aren't around any more when a concurrent study is conducted. The usual effect of attrition is to lower the validity coefficient. Why is that so? Because the people who drop out are usually from one extreme of the criterion distribution or the other (they are either underqualified or overqualified), and this reduces the variance, or range, of scores. We noted in an earlier chapter that reducing the variability of either X or Y reduces the correlation between them.

In certain circumstances, however, it can be dangerous to substitute a concurrent design for a predictive one. Sometimes the predictor might be influenced by, or reactive to, the subsequent experiences of the individual. For example, if success in medical school is to be predicted in part by the ratings made of the candidate by a selection committee after a face-to-face interview, then experience in medical

school may change the individual's response to certain questions. Suppose the candidates were asked about what they thought being a physician would be like and suppose those who initially had realistic expectations do better in medical school. Actual experience in medical school may change the individual's response even though he or she strives mightily to recall what their thoughts were like at the time of application. Another example would be a job-sample test, or simulation, for which actual experience on the job would yield a higher score. Thus if the predictor is reactive to experience, one should substitute a concurrent design for a predictive one with extreme caution. Keep in mind that there are also many predictors that are not reactive to experience. Obviously, in the above example, undergraduate GPA would not be reactive to experience in medical school nor would scores on a general intelligence test. General intelligence is conceptualized as a relatively stable characteristic of people that does not change much over the adult years.

With the above complexities in mind, remember that while a concurrent design might sometimes reasonably be used as a substitute for a predictive design, the coefficient being *estimated* is still a predictive validity coefficient.

Postdictive Validity

Finally, sometimes we are interested in the accuracy with which we can estimate the extent to which an individual possessed a given trait in the past; this is termed *postdictive validity*. In other words, we wish to estimate a criterion score that describes certain characteristics he or she manifested at an earlier time. Why not get these criterion scores directly? Because it may be too difficult, expensive, or time-consuming to do this, and furthermore such records may not be available for all individuals.

For example, suppose we wish to know whether an individual has ever been arrested for a traffic violation. We would have to make inquiries of the appropriate agencies in every state where the person in question has driven a car. This would be quite time-consuming and probably not very easy. A much simpler and quicker approach would be to ask the individual the question: "Have you ever been arrested for a traffic violation?" In response to this question we might expect some falsification and errors of memory, so that answers might not correspond exactly to past history. Nevertheless, if it were found that responses to the question were substantially related to past history, we might well be willing to substitute the question (predictor) for the more time-consuming direct approach (criterion).

Summary Comment on Criterion-Related Validity

Keep firmly in mind that a given test may be used for more than one type of prediction. For example, an arithmetic test may have moderate validity in predicting success in clerical occupations, good validity in the diagnosis of failure in mathematics courses, and some but low validity in predicting whether or not the individual had formal schooling. Similarly, for any one given kind of prediction, a given test may have differing validity. A test of ability to visualize spatial relations may have low validity in predicting success in clerical occupations, moderate validity in predicting success in engineering school, and good validity in predicting success in a machinist's apprentice training course. One cannot separate questions about validity from a consideration of the specific purposes for which the predictor is to be used.

Content Validity

The *content validity* of a set of measurement operations refers to the degree to which those operations measure the characteristics we wish to measure, as *judged* from the appropriateness of the *content* of those operations. We examine a test or other measuring instrument and then, on the basis of our experience or expertise, judge its validity. The question asked is: "To what extent do the operations measure what they are supposed to measure?" The answer is based on professional judgment.

When we recall the many different situations in which we obtain quantitative descriptions of individuals on one trait or another, it would appear that content validity is the type of validity with which we are most frequently concerned. Consider, for example, classroom examinations, thousands of which must be constructed and administered to students every week. They are taken to be valid measures of knowledge of school subjects because the questions they contain all pertain to the subject matter and are drawn from the class discussions, the lectures, and the textbooks. Similarly, in the laboratory we use a wide variety of devices to measure different motor, perceptual, and cognitive abilities. These devices are taken to be valid because the behavior they elicit seems so obviously to reflect the pertinent traits. What else could be measured by an aesthesiometer but touch sensitivity, by a stereoscope but binocular vision, by a maze but learning ability? Indeed a very large proportion of the measuring devices we use seem by their very nature to measure the intended traits.

Determining Content Validity

The nature of content validity and the processes involved in determining it will be clearer if we consider a few examples. We might conclude that a test of simple reaction time is highly valid as a measure of simple reaction time because from our analysis of the situation it is apparent that the individual is not required to make complex decisions, the intensity of the stimulus is considerably above threshold, and the response required is merely to depress a key with the forefinger. In similar fashion, on the basis of our examination of the questions comprising a test of knowledge of U.S. history, we might conclude that the test has only moderate content validity because even though all the questions in it require knowledge of U.S. history, none of them deal with the Civil War period. As another example, we might say that a rating scale designed to measure artistic creativity is a very poor measure of this characteristic because we find that the teachers using it base their ratings almost entirely on the children's ability to draw realistic pictures.

From a study of the content or nature of the operations, then, we arrive at a judgment of the extent to which those operations measure the characteristics we intend them to measure. There are, in fact, two judgments involved: the extent to which each element or item of the test pertains to the variable of interest as it is defined, and the extent to which the entire set of elements or items represents all aspects of the designated domain.

The grounds for judging the extent to which a test item measures the desired properties are not always entirely clear and could easily be incorrect. For example, consider the question: "Would you rather read a book or go to a movie with a friend?" This might be included as a question in an inventory designed to measure the trait of sociability because it involves the choice between a solitary and a social activity. However, it may be that people respond to it on the basis of their interest in literature rather than on their gregariousness. Or it may be that in answering the question, the individual tends to give the "sociable" response as a result of a need to conform and not as a result of a desire to be with others.

Nevertheless, in many instances the components of a test so manifestly reflect the trait that the test is intended to measure, and the situation in which the test is administered so clearly minimizes the effects of other factors, that the judgments about validity seem quite reasonable. For example, to correctly answer the following question in a vocabulary test, the individual must understand the meaning of the key word and of the alternative words.

Which of the following words means the same as *garish*?
 gaudy tarnished irregular rude

Other types of knowledge would be of no value. Therefore while in some instances there may be good and sufficient reason to question the validity of a test as judged by the nature and content of the operations, in others the traits measured seem so obvious that they can be agreed upon by all or almost all judges.

As noted above, not only do we want to know whether each of the components of a test reflects the domain being measured, but also we want to know whether the components taken as a whole cover it in a representative fashion. Judgments of the validity of a test, then, must take into account the extent to which the elements or items that make it up cover all aspects and facets of the trait. For a test of arithmetic ability to be considered to have a high degree of content validity, it should contain questions involving not only addition, subtraction, and multiplication, but also division. A test of knowledge of English literature obviously would not be representative of the subject if it did not have questions dealing with Shakespeare. The content validity of a measure of proficiency in typing would be found wanting if individuals were evaluated only in terms of their speed, and accuracy of performance were ignored.

We should establish the content validity of a test, then, by showing that its elements or items are a representative sample of all aspects and facets of the trait as prescribed by the definition. It is therefore apparent that the adequacy with which we can judge the validity of a test from its content is a function of the adequacy of the definition of the trait. The less detailed and complete the definition, the less well can we judge how representative its components are of the entire trait. When we have a precise definition of a trait, we have a good idea of its limits and we have reasonable knowledge about the nature of its parts. Thus the trait of eye-hand coordination might be defined thus:

Eye-hand coordination is the ability to make quick and accurate movements guided solely by information obtained through vision. This ability involves moving an object by means of the hand to a fixed target or maintaining an object on a moving target. Eye-hand coordination is not just simple speed or accuracy of response or visual acuity, but the combination of these two abilities in the performance of movements relative to a target.

We may not like this particular definition, believing that it covers too much or not enough; but it is reasonably specific, indicates what is included in the trait and what is not, and describes the various areas covered by it.

On the basis of this definition we should know that a valid test of

eye-hand coordination must involve movements of the hand relative to both stationary and moving targets. A test involving just one or the other type of target would not be representative. Similarly, the quantitative description of the individual's performance, his or her score, must reflect both the speed and the accuracy of his or her movements. Furthermore, if the test provides clues to the location of the target through sense modalities other than vision, its validity is impaired.

Unfortunately, even in seemingly clear-cut cases it cannot always be held that equal representativeness of all parts of the domain is sufficient for content validity. Thus it might be argued that the events that took place in American history from 1776 to 1780 shaped the course of the United States far more than those that took place from 1800 to 1840. Hence to be valid, a test of knowledge of American history should have five times as many items dealing with the shorter early period than with the longer later period. In the same way, a mathematician might argue that since the process of division merely involves multiplication and subtraction in an arithmetic test, there should be far fewer division items than those of the other three types. Unless the definition of the trait specifies the relative importance of the various aspects of phases of a trait, subjective judgment is required to determine whether in a test designed to measure that trait they should be equally represented or should have different weights. Again subjective judgment is involved in all phases of content validity and is its paramount characteristic.

Enhancing Content Validity

Depending on the circumstances and the characteristics to be measured there are a number of ways that expert judgments about content validity can be augmented. These are in no way a replacement for subjective judgments. They are merely some possible strategies for enhancing it.

Item homogeneity One approach to enhancing subjective judgment and giving a more precise statement of the degree of validity is to view the relative homogeneity of the components of the test as an indication of content validity. The argument is that if the component parts of the test are intended to measure the same trait, their scores should be positively correlated; and the higher their intercorrelations, the more content-valid is the test. Following this notion, the total score over all the items, the pooled judgment as it were, is viewed as the best available measure of the characteristic that the test is designed to measure. The next step is to eliminate those items that are unrelated to the total test score or that have only a low relationship with it and to retain those items that have high correlations with the total score. To do this, an

item analysis, such as that described in Chapters 7 and 13, is performed in which the correlations between scores on the items and total scores are computed.

By the above process a test can be developed that has greater homogeneity than the original test because the relationships among the items are higher. Thus in addition to the judgmental process by means of which the original test elements were selected or developed, we have added homogeneity as a basis for gauging validity. This is a somewhat different notion from the basic concept of content validity, and has theoretical implications akin to those of the concept of domain sampling described in Chapter 8. Content validity in and of itself depends on subjective judgment alone. Nevertheless, even if we consider the homogeneity of the elements of a test as an "aid" to judgment, it pertains only to that aspect of content validity that is concerned with the intentional sampling of homogeneous content elements. It does not deal with the problem of the extent to which the elements of a test cover all aspects and phases of the trait or characteristic. Indeed, by discarding those elements, we may eliminate an important aspect or phase of the variable. For example, suppose mechanical aptitude is defined as knowledge of mechanical principles, ability to perceive spatial relationships, and manual dexterity. It is likely that those test elements pertaining to the first two aspects of mechanical aptitude would be positively related to a substantial degree, while those pertaining to the last aspect would be unrelated to the other two. If on these grounds the manual-dexterity elements were eliminated, an important aspect of mechanical aptitude *as defined* would not be measured.

Discarding parts of a test to ensure that the remaining elements are homogeneous, then, may result in a new test of lower content validity—lower validity in the sense that it does not measure the trait as defined as well as did the original, heterogeneous test. What we have now is a test that measures only certain aspects of the trait. This new test may be more useful and perhaps will give us new and fruitful ideas about the nature of human abilities, but it does not measure the trait as it was originally defined.

The overall message here is that content validity can be enhanced by selecting homogeneous items only when the definition of the variable specifies that homogeneity is desirable.

Parallel-panels method Another procedure that could enhance our confidence in judgments of content validity is for two or more panels of experts to go through the content-validity procedure independently and to compare the final product. That is, without knowledge of what the others were doing, each panel would study the definition of the domain, generate a pool of possible items, refine the item pool by judging the

extent to which each item is representative of the domain of interest and the extent to which the item pool adequately samples all relevant parts of the domain.

If the measures produced by these independent panels of experts are similar in content, and in fact scores on the measures are highly correlated (as in parallel forms reliability), then the content validity of the measure is enhanced.

Multiple judges It is perhaps too obvious to mention that confidence in content-validity assessments would be enhanced if multiple judges were used and all necessary areas of expertise were represented. That is, if the objective is to build a comprehensive graduate-level test of knowledge of chemistry, many varieties of chemists (for example, organic, physical) would have to be included in the expert panel. It would also be wise to include more than one chemist of each kind. Statistically speaking, the more experts there are, the more the various errors in judgment can be averaged out. Further, at some point it is wise to include a sample of potential users as judges. Often the test takers themselves can spot validation errors that the content experts miss.

Cautions About Content Validity

It is perhaps the case that establishing content validity is especially difficult in those situations where it is needed most. A prime example is in the development of predictors for job performance. The desire to make accurate and fair employment-selection decisions is intense, and it results in considerable pressure to leave no stone unturned in the validation process. Unfortunately job performance is an exceedingly complex domain to sample for purposes of establishing the content validity of predictors, and it is difficult to represent tasks in the selection tests precisely as they occur on the job. The content fidelity of the predictor cannot be perfect even for tasks such as typing, since the office (performance) situation is not quite the same as the test situation. Thus one must use the content-validity strategy with caution. If there is not clear agreement among experts that a sample of measurement operations is content-valid, then additional strategies must be used. In general, these fall under the rubric of construct validity, and they are the topic of the next section.

Finally Guion (1978) has cogently pointed out that merely obtaining a sample of content that is judged to have high content validity is not the end of the battle. An individual's performance *on the test* must still be *scored* so as to yield some indication of whether the individual ranks high, low, or in between on the characteristic(s) measured by the test.

Do the individual differences that appear across scores on a content-valid work-sample test reflect the individual differences that appear across people who are actually performing on the job? The moral of this story is that we must worry about whether the content sample represents the population of content and whether the scoring system for individual proficiency on the sample is correspondent with what it takes to score high or low in the population of content. For example, adequate performance on a work-sample test for an auto mechanic may be assessed by one judge employed by the personnel department to test applicants. However, there are multiple judges on the job (including customers) and these on-the-job appraisal processes may not be the same as the scoring procedures used for applicants.

In summary it might be wise to follow Guion's (1978) lead and speak of content-oriented test development rather than content validity. It really makes no sense to introduce the expert judgments only for assessing the adequacy of the sample of test components after they have been selected. Such judgments should be an integral part of working out the definition of what's to be measured, generating the content sample, and developing the scoring system. Thus content validity really is a label for the complex process of test, or predictor, construction.

Construct Validity

The term *construct validity* (Cronbach and Meehl, 1954) was invented to describe a validation process that was often used but that had never been identified with a label. Such a process takes place whenever the primary objective is to develop a measure of an individual characteristic for its own sake, and we want to know how well the test, or measurement operations, measure this characteristic. That is, the principal focus is not on the prediction of a criterion or on the one-to-one correspondence between the content of the test and a specific domain of knowledge or behavior, but on the ability of the test itself to measure the individual trait or characteristic of interest.

Most often such questions arise when someone thinks it is important to develop a measure of a psychological property of individuals for which, in fact, there is no existing, real-world measure (criterion). Mental abilities such as general intelligence, mechanical aptitude, and spatial relations; and personality characteristics such as sociability, dominance, the need for achievement, and introversion fall in this category. Since such things are hypothesized to exist, but have no direct real-world counterpart (That's why they are *hypothesized* to exist!), the variables in question are called *constructs*, and hence the term construct

validity. Our interest is in developing a test that actually measures such things because having such a measure would be important in a broad range of practical and theoretical situations. For example, clinicians and counselors want to know something about their clients' personality characteristics so as to more intelligently explore how a client might react to a variety of future situations. Motivation researchers might need a good measure of need achievement because it is necessary before they can test their favorite theory in a controlled experiment (for example, people scoring high on the need for achievement will act one way; people low on the need for achievement will act another). Researchers interested in the question of genetic versus environmental control of general intelligence really need a good measure of intelligence before they can proceed. In sum, such characteristics are referred to as constructs since they are hypothesized to exist but there is no one criterion measure for them. Such constructs are all around us and are not limited to personality factors and mental abilities. For example, in athletics there is a construct called "endurance" that is frequently talked about. Trying to be more systematic Fleishman (1975) identified five underlying "strength" constructs (for example, static strength, dynamic strength) that he argued would explain performance on a wide range of specific motor tasks (pushups, the 100-meter dash, distance swimming, and the like). When leadership behavior is studied, constructs such as "consideration for subordinates" (giving praise, explaining reasons for action, asking opinions) and "initiating structure" (setting goals, keeping on schedule) are used, and searching questions are asked about how validly these constructs are measured by the questionnaires that were designed for them. In sum then construct validity asks the question: How well does a set of operations (that is, a test) measure an individual's standing on a construct?

We are not asking how valid the test is for predicting a particular criterion. When we concern ourselves with predictive validity, we are in effect saying that we have no real interest as such in the operations that form the predictor, but only in the criterion. Criterion scores are precisely what we want, but because of the force of circumstances we cannot obtain them directly. Therefore, we seek some other operations that we can use to obtain estimates of criterion scores. However, in general, this occurs only in those limited practical situations where we wish to forecast some future behavior such as success in school or on a job. More often our interests are broader. For theoretical purposes we are concerned with the relationships among psychological traits that are intellectual constructs and with the effects of various conditions on such traits. To be sure, we may be interested in the predictive value of the traits, asking such questions as the extent to which measures of intelligence forecast later success in school. However, in such in-

stances of specific decisionmaking, our interest clearly shifts to a concern for the operations as predictors and not as a means for assessing an individual's standing on a construct.

In many instances, while predictive validity may seem to be the primary focus, in fact it is not. For example, if we wish to establish the validity of a test of scholastic aptitude, we might do it by ascertaining the correlation between test scores and grades earned in college. But what college, what major, and for what period of time? The power of a test to predict grades may appear to differ from one college to another, from one major to another within the same college, and from one time period to another. We might explain away these variations in relationship by saying that grades in some colleges are more reliable than in others and hence more predictable, that in some colleges grades are based less on academic achievement than on other factors, and that some college majors depend more heavily on cognitive ability factors than do other majors. We also might say that in the past almost any high school student was admitted to college, where as now there is a considerable restriction in the range of ability of those who are admitted; or that the high school preparation of students in the past was better or poorer than that at present. In other words, we are taking the point of view that our test measures a construct called scholastic aptitude, but that there is never a single satisfactory criterion available for the evaluation of its validity. To put it another way, we are saying that there is a trait of scholastic aptitude that is measured by our test, and we offer as evidence the fact that test scores are positively related to success in a variety of different academic situations. Because these correlations are not perfect, we conclude that academic success is determined by factors other than scholastic aptitude. Indeed we may be able to demonstrate this by showing that among individuals with the same scholastic aptitude, that is, with the same score on our test, those who are highly motivated earn higher grades than do those who are less motivated.

All of this is not to deny the legitimate role of criterion-related validity. Indeed there are many instances where it is of primary concern. Rather the point is that in many situations predictive validity alone cannot tell us what we want to know about a measure.

Describing Construct Validity

The establishment of construct validity must begin with the theoretical definition for the construct. Most likely the construct is embedded in some theory about behavior, and the theory specifies in some fashion or other (1) the meaning of the construct, (2) how it is related to other constructs, and (3) how it is related to specific observable behaviors. For example, we might be interested in a construct called "garden-

variety anxiety." Let's suppose that this type of anxiety is not the deep-seated psychopathological kind that characterizes people for long periods of time, leads to suicide, can be relieved only by powerful drugs, and so on. Neither is it the kind of momentary anxiety you feel as the result of being caught in a blizzard or spilling gravy on yourself in a restaurant. Rather it is the kind that you experience as the result of anticipating fairly major events in your life such as entrance exams for law school, the possibility of a divorce, or losing your job. Such anxiety would be expected to fluctuate over a period of months as major circumstances in your life change, and *some people would react more strongly to such changes than others*. This kind of anxiety is *not* related to general intelligence (says our theory), but is negatively related to self-esteem. Further there is a curvilinear (inverted U-shaped) relationship between garden-variety anxiety and performance on course examinations. That is, if you want to do well, it is best to have a moderate amount of such anxiety.

If we were really serious about investigating this construct, we would have to be much more thorough about defining it and mapping out its hypothesized relationships with specific behaviors and other constructs. The above brief discussion is meant to be illustrative only.

Once a definition of the construct is available, a set of measurement operations must be developed, using the methods that are discussed elsewhere in this book. For example, we might construct a paper-and-pencil test of garden-variety anxiety.

One major confusion regarding construct validity at this point concerns the breadth of its interpretation. In its widest sense one could think of construct validity as pertinent to any theory of behavior. That is, understanding the construct, or knowing the theory, permits a host of predictions about behavior when the construct is used as the independent variable. In this sense, construct validation is a *deductive* process, and construct validity is established to the extent that specific hypotheses deduced from the theory are substantiated in empirical studies. By contrast, an investigator could be concerned with giving substantive meaning to a particular set of measurement operations (for example, a specific paper-and-pencil inventory) for which he or she was not yet ready to predict empirical relationships. Thus the process becomes *inductive*, and what's being validated is not a theory about some characteristic of individual differences, but the content of a particular test. Consequently, construct validity would be established to the extent that a large number of relationships between the test and other variables was determined, and the pattern of relationships that was found clearly indicated the meaning of the test score.

The original APA Technical Recommendations made construct validity sound almost synonymous with the standard ingredients of deductive theory building, which detracted somewhat from its uniqueness in

the validity taxonomy. The 1966 statement of test standards moved away from this position a bit, but remained committed to the deductive view. In our opinion, taking either extreme is not in the best interests of psychometric theory. Viewing construct validity as purely deductive theory building gives it no distinctive reason for being. Also the psychological study of individual differences is still at a primitive enough level that many promising avenues of research could be dried up prematurely if a deductive approach were required and the first few hypotheses based on the theory about the construct went down the drain. It would seem more productive for a validity model to give investigators some guidance for a certain amount of inductive "fishing" before moving to substantive hypothesis testing. At the other extreme, adopting a strictly inductive view throws away the power of hypothesis testing. If investigators never stick their necks out with regard to what they think their instruments are measuring, they will never establish surplus meaning or rule out competing explanations as to why a measure acts the way it does in the empirical world.

A second important distinction regarding construct validity is whether the theory being validated concerns the construct as it is represented by a specific set of measurement operations or whether it is independent of any specific measuring instrument. The latter option again makes construct validity synonymous with the validity of almost any psychological theory and detracts from its unique character. The power of the construct-validity notion is derived from its function as a means for assessing the validity of a specific measuring instrument intended to measure a human characteristic for which there is no naturally occurring criterion.

Determining Construct Validity

Again construct validity applies when the investigator wants a measure of a characteristic that is deemed important by somebody, but for which there is no already available indicator. The "thing" being validated is a set of measurement operations, and the process should include both an inductive and a deductive component. The following kinds of studies make useful contributions to construct validity:

1. The theory (definition) of the construct should say something about how the individual measurement operations (for example, the items in a test) should be interrelated. If the test is designed to measure only one factor, then the individual items should all intercorrelate positively. If, for example, our theory of garden-variety anxiety says that we have anxieties about

social situations and anxieties about achievement situations, then the intercorrelations should reveal two groups of items. The items referring to social situations should correlate highly among themselves and so should the items referring to achievement situations. However, the correlations between the two types of items should not be as high. In general, the clustering of the item intercorrelations should be consistent with the nature of the factors specified in the construct definition.

2. Given the definition of the construct, we might expect some reliability estimates to be high and some to be low. For garden-variety anxiety, the parallel-forms or test-retest method over a several-month period may yield rather low estimates, while the split-half estimate should be high.
3. The measure of the construct certainly can be correlated with many other measures. If the theory says that some of these measures reflect the same construct, then these correlations should be high, and vice versa.
4. Correlating the measure of the construct with other variables that might account for, or rule out, potential sources of bias or irrelevant variance in the instrument being validated is also useful. For garden-variety anxiety such sources of nonvalid variance might be associated with substantive content (say, verbal fluency) or methodological elements (item format, social desirability). For example, it would be good to know if responses to our paper-and-pencil anxiety measure were related to vocabulary proficiency. We wouldn't want that. Also, if responses to the anxiety test were too highly related to measures of social desirability, then we would try to modify the anxiety test to get rid of that extraneous source of individual differences.
5. A special kind of correlational evidence is provided by the *multitrait, multimethod matrix (MTMM)*. Such a matrix can be generated when we have at least two constructs being measured by at least two different methods (Campbell and Fiske, 1959). Consider the simple example of an MTMM matrix in Table 10-1, where there are two traits, *A* and *B* (for example, anxiety and sociability), and two methods (paper-and-pencil inventory and observer ratings of behavior).

It follows that if we understand what we are measuring, two different measures of the same thing should intercorrelate highly. Campbell and Fiske called this *convergent validity*. A more stringent consideration is to stipulate that if we under-

Table 10-1 Example of a multitrait, multimethod matrix (MTMM)

	Trait A		Trait B	
	Method 1	Method 2	Method 1	Method 2
Trait A	Method 1	$r_{1A,1A}$ $r_{1A,2A}$	$r_{1A,1B}$	$r_{1A,2B}$
	Method 2	$r_{2A,2A}$	$r_{2A,1B}$	$r_{2A,2B}$
Trait B	Method 1		$r_{1B,1B}$	$r_{1B,2B}$
	Method 2			$r_{2B,2B}$

stand what we are doing, the correlation between two methods designed to measure the same trait (for example, $r_{1A,2A}$) should be substantially higher than the correlation between two traits when they are measured with the same method ($r_{1A,1B}$). This stipulation is called *divergent validity*, and it has humbled many an investigator. The correlation between two variables should not be a function of similarity in measurement method. It should be a function of similarity in substantive content.

We should note one important caution at this point. The MTMM strategy really assumes that the two methods have been constructed with equal skill. If one or the other is shortchanged (for example, if any old method of quick and dirty observation is compared with a carefully developed paper-and-pencil inventory), it most likely will be unreliable, therefore not correlate with anything, and neither convergent nor divergent validity will be observed. Proceed with caution here.

- Valuable sources of information on construct validity are experimental studies using the measure being validated either as the dependent variable or as a cross-classification variable. For example, to investigate garden-variety anxiety, we could (if we were mean enough) divide a college class randomly into two groups at the beginning of the term. One group could be told that only 10 percent of the class will get an A and 50 percent will fail, while the second group could be told that no one will fail and at least 50 percent will get an A. The mean score on our anxiety test for these two groups *should* be different after this experimental treatment is imposed. In general, there should be any number of experiments for which we can predict changes on our construct measure *if* we understand it.
- Content validity becomes a part of construct validity if and when we ask experts to judge the adequacy of the methods

used to sample item content from the universe of content, and the adequacy with which the measurement operations seem to reflect the theoretical definition of the construct.

- A final line of evidence is derived from what we will call a *process analysis* of individuals responding to a construct measure. This would entail such things as the detailed questioning of subjects as to why they responded a certain way to a test item or the questioning of raters as to how and why they actually chose a particular rating for an individual. The subjects, or raters, should be using a process reasonably close to that which the researcher had in mind when he or she developed the instrument. Such process data is often an eye-opener.

Almost all of these different lines of evidence can be used in both an inductive and a deductive manner. At some point, however, things must get serious and the investigator must stick his or her neck out and *predict* how the instrument will work in empirical studies. Two principal characteristics make one type of hypothesis better than another. First, more construct validity can be attributed to the instrument to the extent that the hypotheses being tested are nonoverlapping. That is, much less is gained from confirming the same hypothesis ten times than from confirming ten hypotheses of very different kinds. If we are validating a measure of garden-variety anxiety, we could predict its correlation with grade-point average in ten different random samples from the same population, or we could make ten different predictions, such as the responsiveness of the anxiety scores to threats of low grades, the kinds of reliability that would be high and low, the correlation of the anxiety test with intelligence, and so on. Second, more construct validity is accumulated to the extent that hypotheses are more stringent and are easier to negate. For example, it is not very stringent—and we don't learn very much—if we support the hypothesis that under conditions *X*, *Y*, and *Z*, garden-variety anxiety correlates with job performance somewhere between $-.60$ and $+.60$. It would mean a great deal more to confirm a prediction that these two variables are correlated between $+.30$ and $+.40$.

If construct validity is viewed in the above fashion, it does have a unique place in psychometric theory. However, we should keep in mind that it does not come cheaply.

Methodological and Statistical Problems in Criterion-Related Validity

There are a number of methodological and statistical problems that arise in connection with criterion-related validity. Because criterion-

related validity plays a part in the determination of construct validity, these matters are also pertinent to it. In this section we discuss the prediction of criterion scores, reliability as it relates to validity, prediction when there are multiple criteria, and validity in relation to the range of individual differences.

Estimating Criterion from Predictor Scores

In our discussion of correlation and regression in Chapters 5 and 6 we learned that if we know an individual's predictor score X , our best estimate of that individual's criterion score is the average of the criterion scores, $\bar{Y}_i$, of all those individuals in the sample who have the same predictor score. If we take the relationship between predictor and criterion scores to be linear, then we can express the degree of relationship by means of the Pearsonian coefficient and, it will also be recalled from Chapters 5 and 6, we can then use the regression equation to give us a value we take as representative of the Y scores of individuals all of whom earn the same X score. That is, when we have a single predictor, we use the regression line (the straight line of best fit in a least-squares sense); and when we have several predictors, we use the regression plane (the plane of best fit in a least-squares sense) to make our prediction from X to Y .

The accuracy with which these predictions are made is described by the standard error of estimate or prediction (Equation 6-19 when the predictor is a single variable and Equation 11-25—discussed in the next chapter—when it is a composite variable). The standard error of estimate is the standard deviation of the criterion scores of those individuals all of whom have the same predictor score. It should be recalled from our discussion in Chapters 5 and 6 that when the relationship is taken as being linear and is described by the Pearsonian coefficient, the deviations on which the standard error are based are deviations from the regression line and not from the means of the Y scores in the columns, unless those means happen to fall on the regression line. The standard error of estimate or prediction, then, describes the accuracy with which individual predictions are made by the regression equation. It will be remembered that we must adopt either of two points of view about the standard error of estimate. If we are willing to presume the relationship is homoscedastic, then we can take the standard error as descriptive of the accuracy of predictions at all levels of predictor scores. If we are not willing to do this, we must take the standard error as a weighted average of the errors of prediction.

Index of Forecasting Efficiency

Of some historical interest is another index of how well criterion scores can be estimated from predictor scores; it is called the *index of forecasting efficiency* (E). Since the objective is to represent the proportional decrease in the standard deviation of the predicted scores when the new predictor is compared to no information (that is, predict the overall mean of Y for everybody), the index is defined as

$$E = \frac{\sigma_y - \sigma_y \sqrt{1 - r_{xy}^2}}{\sigma_y} \quad (10-1)$$

$$= 1 - \sqrt{1 - r_{xy}^2}$$

The second term in the numerator of the first expression is obviously the standard error of estimate using the predictor.

Effects of Reliability

Equation 9-6 shows how the reliability with which two variables are measured affects the correlation between them. This formula rearranged is

$$r_{xy} = r_{xxyx} \sqrt{r_{xx} r_{yy}}$$

That is, given r_{xxyx} (the correlation between true predictor and true criterion scores), we find that as r_{xx} (the reliability of the predictor) and r_{yy} (the reliability of the criterion) change, so does the empirically determined validity coefficient, r_{xy} . We can see that as the reliability of either the predictor or the criterion becomes lower and lower, the validity becomes lower and lower. As we saw in Table 9-1, if either the predictor or the criterion is completely unreliable, then fallible scores on the two variables will be completely uncorrelated and the validity will be zero. Hence we can say that reliability limits validity and that for optimal prediction both the predictor and the criterion should be measured with as high reliability as possible, so that r_{xy} approaches r_{xxyx} , the highest possible validity.

Correcting for Attenuation

If the validity coefficient is limited by the lack of predictor and/or criterion reliability, it is said to be attenuated. As noted in Chapter 9, if

we keep certain precautions in mind, we can forecast what validity would be if the predictor and/or the criterion were measured with perfect reliability. That is, we can correct for the attenuation.

From the discussion in Chapter 9 and by rearranging the above formula, we can see that if the reliability of the predictor X were perfect, then the correlation between a predictor with perfect reliability and a fallible criterion would be

$$r_{xy} = \frac{r_{xy}}{\sqrt{r_{xx}}} \quad (10-2)$$

Similarly, if only the criterion were corrected for attenuation, then the forecast of the correlation between a fallible predictor and a criterion with perfect reliability would be

$$r_{xy} = \frac{r_{xy}}{\sqrt{r_{yy}}} \quad (10-3)$$

Finally, if both predictor and criterion were corrected for attenuation, then the correlation between a perfectly reliable predictor and a perfectly reliable criterion would be

$$r_{xy} = \frac{r_{xy}}{\sqrt{r_{xx}r_{yy}}} \quad (10-4)$$

Again, in line with the discussion in Chapter 8, the estimates of reliability that are used in these corrections must be appropriate ones. For example, if true scores change over time, then a test-retest or equivalent-forms coefficient would count such changes as error, or unreliability, and the reliability estimate would be artificially low. If the reliability estimate were artificially low then the corrections for attenuation given by the above formulas would be artificially high. The same reasoning would apply if an internal-consistency method (for example, coefficient alpha) were used to estimate the reliability of a multidimensional measure.

Given these cautions, it is often argued that when criterion-related validity is estimated, the correlation between a predictor and a criterion of perfect reliability should also be estimated; that is, the validity coefficient is corrected for attenuation in the criterion. For example, if the validity coefficient is .35 and the reliability of the criterion (for example, a measure of job performance) is .49, the "corrected" validity coefficient is $.35 \div \sqrt{.49}$ or .50. Which number is the better estimate of what the validity of the predictor will be with future samples of people from this same population? If the reliability coefficient of .49 is an appropri-

ate one, and is therefore an indicator of the unsystematic random errors of measurement, then .50 is the better estimate of validity. After all, the individual's actual contributions to the organization are not limited by the errors inherent in the method used to measure the criterion. Defined as we are defining them, there is no measurement error in *actual* contributions.

However, if the reliability coefficient used to make the correction is not an appropriate one, in the context discussed above, all we can say is that the fallible-score coefficient is too conservative, but we don't know to what extent.

Finally, when considering criterion-related validity, it is generally not appropriate to correct for unreliability in the predictor. In applied psychology we must always deal with fallible predictor scores. That is, the decision must be made on the basis of the predictor data in hand, and it is the validity of the fallible predictor information that is at issue. For example, it matters not if near-perfect predictor reliability could be achieved by increasing the length of the test to 3,000 items. The "best" sample of 100 items may be all we can get, and it is the reliability of those 100 items with which we must live.

Prediction with Multiple Criteria

We have been discussing predictive validity as if the criterion were always a single variable. However, as is the case with other measures of psychological characteristics, the criterion often is a composite variable. A criterion of leadership might be a series of rating scales designed to measure various facets of that domain. The criterion of success for a particular sales job might involve the total dollar volume of sales, the number of new accounts developed, and the number of repeat sales to the same customers. In cases such as these, when we have multiple criteria, we must somehow combine them in order to compute the coefficient of correlation between criterion and predictor scores so as to describe predictive validity.

In combining criteria, we can take all of them to be of equal importance or we can differentially weight them. Criteria can be taken to be of equal importance either in the sense that the units of measurement in which each is expressed are equivalent, so that they can be summed or averaged, or in the sense that they contribute equally to the variation in composite scores. If we wish to have the various component criteria contribute differently to the composite, they can be assigned differential nominal weights based on personal judgment about their relative importance, their reliability, or their correlation with a principal construct that underlies all the component criteria.

Finally, critical points may be set on each criterion and individuals placed into various classes, depending on the number or pattern of criterion cutoff points they exceed.

Assuming that multiple criteria are equally important In many instances the scales on which the different criteria are expressed are assumed to be equivalent, and scores on all component criteria are assumed to be comparable and therefore simply summed or averaged. Thus when academic success in college is measured by grades received in different courses, all these grades are taken as being scores on comparable scales and are averaged to obtain a total criterion score. This is also likely to be done with ratings, where ratings on a series of scales designed to measure different facets of a trait are merely summed or averaged. In some cases it is possible to change the units in which the various scales are expressed to other units that are equivalent. For example, suppose the success of production employees is gauged both by the number of items of work they produce and by the number of accidents in which they are involved. If the value of an item of work is known, production can be expressed in terms of dollars. Similarly, if the cost of an accident is known, it too can be expressed in terms of dollars. The composite criterion score for an individual would then be the dollar value of production minus the dollar cost of accidents. The net value would be the individual's contribution to his or her job.

If we wish to consider the various criteria as being of equal importance in the sense that they contribute equally to the variation in composite scores, the common procedure is to transmute the scores on each of the component criteria to standard or standardized scores. However, as we saw in Chapter 7, this ignores the intercorrelations among the criteria, and these intercorrelations also contribute to the variation in total scores. Therefore transmuting scores on each component criterion to standard or standardized scores does not result in a situation wherein each contributes equally to the variation in composite scores, though this result may be approximated thereby. But as we saw in Chapter 4, the use of standard or standardized scores does in a sense make the units comparable.

Differential weighting of component criteria When it is believed that the various component criteria should be differentially weighted, integer weights can be assigned and differentially weighted composite scores obtained. Ordinarily differential weights are applied to component criterion scores after the scales have been made comparable, by means of standard or standardized scores. Either we can use weights that in our personal judgment best describe the relative importance of the component criteria, or we can differentially weight them in terms of their

reliability of their correlations with a construct that we assume underlies all of them.

Perhaps the most common method of developing differential weights is through personal judgment. On the basis of our theoretical notions about the characteristic being measured, our analysis of the characteristic, our values, or similar considerations, we make decisions about the relative importance of the various component criteria. Thus, in gauging the performance of sales personnel, the management of the department store might believe that dollar volume of sales is twice as important as number of absences and four times as important as accuracy in making change. Then the weights of 4, 2, and 1, respectively, would be assigned to the three component criteria.

Seldom does it make sense to weight the various component criteria in terms of their reliability. While it can be argued that the less reliable criteria should not be given as much weight as the more reliable ones, since scores on them are largely determined by unsystematic factors, there is no necessary relationship between reliability and importance. Indeed it may very well be that the least important criterion elements are the easiest to measure. The world does seem to work that way. That is, the things most worth having are the most difficult to get.

In general, the differential weighting of criterion components is, and should be, a value judgment that should be faced directly, and not swept under the rug.

Critical cutoff points In some situations, where the criterion has many different facets each of which is measured by a different component criterion, it may seem that a simple or weighted summation of standard or standardized scores is not appropriate. For example, an airplane pilot whose takeoffs are smooth and whose cross-country flights are precise, but who is unable to land the aircraft without serious consequences, could not be considered a successful pilot. Similarly, a child who is well-mannered at home and in adult groups and whose behavior in the classroom is excellent, but who in play situations is forever hitting, biting, and kicking other children, cannot be considered to be making a satisfactory social adjustment. In these instances as in many others it appears that critical points can be set on the component criteria, below which any score automatically marks the individual as being low on the characteristic the total criterion is designed to measure. A low score on any one component is not taken as being compensated for by high scores on the others as is the case when we average the scores on the component criteria. On the other hand, an individual will be categorized as successful if his or her scores on all criteria fall above the critical points, even though they exceed them by very small amounts. Individuals, then, are placed in one or the other of two cate-

gories, those who are successful or high on the criterion and those who are unsuccessful or low on it. Consequently, we have a discontinuous dichotomous criterion, and validity coefficients are most appropriately expressed by the point biserial or phi coefficient.

The critical points might well be set at different points on the different component criteria, depending on what we consider critical performance on each to be. In this sense this multiple-cutoff procedure does provide a system of differential weights for the component criteria, but because each component is given a complete "veto" power, in another sense they are all equally important.

In our opinion, almost all jobs or educational programs are complex enough to yield multiple criteria of performance that are too distinct to be automatically combined (weighted or unweighted) into a single composite. That is, performance on one critical component is not highly correlated with performance on another. Adding them together hides too much information. However, it is also true that overall decisions such as pass/fail, hire/not hire, promote/not promote must be made about individuals based on all the evidence. In such instances we can only argue, as others have done, that the criterion and predictive validity information should not be combined until the very last minute, and then with the realization that the weight given to each item of information is a necessary value judgment.

Range of Individual Differences

Sometimes we are unable to compute the validity coefficient on the entire range of scores for which the predictor *will actually be used*. The only data available are the predictor and criterion scores for a *restricted range* of individuals. As is the case with the reliability coefficient, or any correlation coefficient, the magnitude of the validity coefficient is affected by the range of individual differences. Consequently, a description of the validity of the predictor based on the scores of the restricted range might not be considered satisfactory, and would generally underestimate the actual validity.

There are two common situations where this circumstance occurs, one involving two variables and the other three variables. In the first situation a restriction of range results from a selection of individuals on the basis of their predictor scores, as would be the case if the predictor were already being used to select people above a certain cutoff score. That is, predictor (X) and criterion (Y) scores are available only for individuals whose predictor scores are above some critical point, and we must estimate the validity of the predictor from this restricted sample of scores alone. There is, then, a *direct* or *explicit* restriction of the

range of individual differences on the basis of X scores, and consequently the standard deviation of X scores is reduced. Any effects of this restriction on the extent of variation in Y scores are *indirect* or *incidental*.

In the second common situation the explicit selection is made on the basis of another variable, Z . Criterion (Y) and predictor (X) scores are available only for individuals whose Z scores fall above a certain critical point, and we must estimate the validity of the predictor from their scores alone. There is, then, explicit restriction on the basis of Z scores, and consequently the standard deviation of Z scores is reduced. Any effects of this restriction on the extent of variation of Y and X scores are *incidental*.

As an example of the first situation, suppose a company uses scores on a clerical test to select office personnel. All those whose scores fall above a given critical point are offered employment, and all those whose scores fall below this point are rejected. As a consequence, test and criterion scores are available only for those persons whose test scores fall above the critical point. Therefore the coefficient of correlation between the test and the criterion of job success describes the accuracy of prediction only for this restricted range of persons and not for the entire range of applicants. Obviously it would be very helpful if we could estimate the validity of the test for the entire range of applicants from that of the restricted range of those who are selected since, in fact, selection of job applicants via the test is from the entire range.

As an example of the second situation, let us say that students have been admitted to college on the basis of their high school grades, with only those whose grades are above a given point being admitted. Suppose a scholastic aptitude test is being considered as a substitute admission procedure. To determine the validity of the test in predicting college grades, it might be administered to entering freshmen. However, these individuals have *already* been selected on the basis of their high school grades, which are most likely correlated with the scholastic aptitude test scores, and as a consequence the correlation between their test scores and college grades would describe the accuracy of prediction only for a restricted range and not for the entire range of applicants. Again it would be helpful if the validity of the test for the entire range could be estimated from its validity over the restricted range.

In the two foregoing examples we have illustrated the restriction of range by the selection of high-scoring individuals. In the following developments, however, no presumptions at all are made about the zone where the scores of the selected individuals fall, but only that their scores do not cover the entire range. It might be that only low-scoring individuals are selected or those in the middle of the distribution, so that individuals at both extremes are excluded.

Given certain assumptions, there are many formulas that can be developed to estimate the validity coefficient and/or the variance of the predictor and/or the criterion in the restricted or in the unrestricted group. Discussing all possible variations of this problem would take too much space, so we present only what seem to be the most common situations. In general, to obtain these estimates, the investigator must know both the restricted and the unrestricted variance for one of the variables (predictor or criterion) and their correlation either in the restricted sample or in the unrestricted sample.

Restriction of range in two-variable situation Let us symbolize the characteristics of the distributions of predictor and criterion scores and their relationships for the restricted and for the unrestricted range as in Table 10-2. We are concerned with the effects of deliberate or explicit restriction of range of predictor scores on the validity coefficient. The ratio $\sigma_x'^2/\sigma_x^2$ describes the amount of explicit restriction on predictor scores, and the ratio $\sigma_y'^2/\sigma_y^2$ the resulting amount of incidental restriction of criterion scores. In each case the amount of restriction is expressed as the ratio of the variation in scores of the restricted group to the variation in scores of the unrestricted, or total, group.

To proceed further, it is necessary to make two assumptions. First we must assume that the slope of the best-fitting regression line through the Y scores in the columns is the same both for the restricted and for the total range of X scores. That is, we assume

$$b'_{y \cdot x} = b_{y \cdot x} \quad (10-5)$$

Second we must assume that the standard error of estimate, or the standard deviation of the Y scores in the columns, is the same both for the restricted and for the total range of X scores. That is, we assume

$$\sigma'_{y \cdot x} = \sigma_{y \cdot x} \quad (10-6)$$

What we are assuming is that the characteristics of that portion of the bivariate distribution, or scatter diagram, that we do have are the same as those of the total bivariate distribution. If the amount of restriction is small, so that σ_x' is almost as large as σ_x and the ratio σ_x'/σ_x approaches 1, the chances are greater that the characteristics of the restricted bivariate distribution will be the same as those of the total distribution.

It is not necessary to assume that the relationship between test and criterion scores is linear in the sense that the means of the Y scores in each column fall precisely on the same straight line and that all the

Table 10-2 Predictor and criterion scores for restricted and unrestricted range

	Unrestricted Range	Restricted Range
Standard deviation of predictor scores	σ_x	σ'_x
Standard deviation of criterion scores	σ_y	σ'_y
Validity coefficient	r_{xy}	r'_{xy}
Regression coefficient	$b_{y \cdot x}$	$b'_{y \cdot x}$
Standard error of estimate or prediction	$\sigma_{y \cdot x}$	$\sigma'_{y \cdot x}$

standard deviations of the Y scores in the columns are equal, so that homoscedasticity holds. In Chapter 5 we saw that the regression coefficient describes the slope of the straight line of best fit in a least-squares sense. All we need to assume is that the best-fitting straight line through that portion of the bivariate distribution we do have is also the best-fitting straight line through the remaining portion. Also in Chapter 5 we saw that $\sigma_{y \cdot x}^2$ can be thought of as the weighted mean of the variances of the Y scores in the columns. Thus it is necessary for us to assume only that the weighted mean of the variances of the Y scores in the columns of that portion of the bivariate distribution we do have is equal to the weighted mean of the variances of the Y scores in the columns of the remaining portion.

From Chapter 5 we can write the regression coefficient as

$$b_{y \cdot x} = \frac{\Sigma xy}{n \sigma_x^2}$$

Since $r_{xy} = \Sigma xy / n \sigma_x \sigma_y$, we can write $r_{xy} \sigma_y = \Sigma xy / n \sigma_x$. Consequently

$$b_{y \cdot x} = r_{xy} \frac{\sigma_y}{\sigma_x}$$

and similarly

$$b'_{y \cdot x} = r'_{xy} \frac{\sigma'_y}{\sigma'_x}$$

Since, by assumption in Equation 10-5, $b'_{y \cdot x} = b_{y \cdot x}$,

$$r'_{xy} \frac{\sigma'_y}{\sigma'_x} = r_{xy} \frac{\sigma_y}{\sigma_x} \quad (10-7)$$

Solving for σ'_y , we have

$$\sigma'_y = \frac{r_{xy}\sigma_y\sigma'_x}{r_{xy}\sigma_x} \quad (10-8)$$

By assumption in Equation 10-6, $\sigma'_{y \cdot x} = \sigma_{y \cdot x}$; so substituting from Equation 6-19, the standard error of estimate, we have

$$\sigma'_y \sqrt{1 - r_{xy}^2} = \sigma_y \sqrt{1 - r_{xy}^2} \quad (10-9)$$

Squaring Equation 10-9 and substituting for σ'_y from Equation 10-8 gives

$$\frac{r_{xy}^2 \sigma_y^2 \sigma_x'^2}{r_{xy}^2 \sigma_x^2} (1 - r_{xy}^2) = \sigma_y^2 (1 - r_{xy}^2)$$

Dividing both sides of the equation by σ_y^2 gives

$$\frac{r_{xy}^2 \sigma_x'^2}{r_{xy}^2 \sigma_x^2} (1 - r_{xy}^2) = 1 - r_{xy}^2 \quad (10-10)$$

Multiplying both sides of the equation by $\sigma_x^2 / r_{xy}^2 \sigma_x'^2$ gives

$$\begin{aligned} \frac{\sigma_x^2}{r_{xy}^2 \sigma_x'^2} \frac{r_{xy}^2 \sigma_x'^2}{r_{xy}^2 \sigma_x^2} (1 - r_{xy}^2) &= \frac{\sigma_x^2}{r_{xy}^2 \sigma_x'^2} (1 - r_{xy}^2) \\ \frac{1 - r_{xy}^2}{r_{xy}^2} &= \frac{\sigma_x^2 (1 - r_{xy}^2)}{r_{xy}^2 \sigma_x'^2} \\ \frac{1}{r_{xy}^2} - 1 &= \frac{\sigma_x^2 (1 - r_{xy}^2)}{r_{xy}^2 \sigma_x'^2} \\ \frac{1}{r_{xy}^2} &= 1 + \frac{\sigma_x^2 (1 - r_{xy}^2)}{r_{xy}^2 \sigma_x'^2} \\ &= \frac{r_{xy}^2 \sigma_x'^2 + \sigma_x^2 (1 - r_{xy}^2)}{r_{xy}^2 \sigma_x'^2 + \sigma_x^2} \end{aligned}$$

Inverting gives

$$\frac{r_{xy}^2}{1} = \frac{r_{xy}^2 \sigma_x'^2}{r_{xy}^2 \sigma_x'^2 + \sigma_x^2 (1 - r_{xy}^2)}$$

Dividing the numerator and denominator of the fraction on the right-hand side of the equation by σ_x^2 and taking the square root of each side gives

$$r'_{xy} = \frac{r_{xy}(\sigma'_x / \sigma_x)}{\sqrt{1 - r_{xy}^2 + r_{xy}^2 (\sigma_x'^2 / \sigma_x^2)}} \quad (10-11)$$

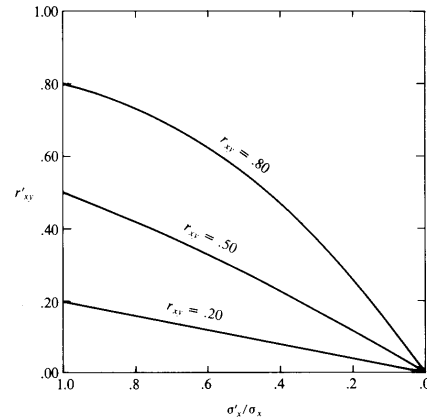


Figure 10-1 Effects of deliberate or explicit restriction of range of predictor (X) scores on validity of predictor (r'_{xy}).

Equation 10-11 shows what the validity of a predictor, r'_{xy} , would be if we deliberately reduced the variation in X scores from σ_x to σ'_x . To show these effects, Equation 10-11 is plotted in Figure 10-1. In this figure we can see that as the variation in predictor scores is decreased, that is, as σ'_x / σ_x becomes smaller, the validity of the test, r'_{xy} , also becomes smaller. It is therefore apparent that if a test is validated on a group whose predictor scores do not represent the total range, the validity coefficient will be underestimated.

Equation 10-11 shows that the validity coefficient will be reduced when the range of predictor scores is reduced. Ordinarily we have data on a group with a restricted range and we wish to estimate what the validity coefficient would be if we had the total range. That is, we want to estimate r_{xy} rather than r'_{xy} . To obtain this value, we start with Equation 10-11, solving for r_{xy} . The solution is parallel to that for r'_{xy} , so we can write

$$r_{xy} = \frac{r'_{xy}(\sigma_x / \sigma'_x)}{\sqrt{1 - r_{xy}^2 + r_{xy}^2 (\sigma_x^2 / \sigma_x'^2)}} \quad (10-12)$$

Suppose that individuals are selected for an office job on the basis of their scores on a clerical aptitude test, the standard deviation of the scores of all the applicants being 10 and the standard deviation of those

selected being 5. For those persons who are hired, we find the coefficient of correlation between test and criterion scores to be .30. From Equation 10-12 we can estimate the validity the test would have if all persons had been hired and there were no restriction in range. Placing the appropriate values in Equation 10-12, we have

$$r_{xy} = \frac{(.30) \frac{10}{5}}{\sqrt{1 - (.30)^2 + (.30)^2[(10)^2/(5)^2]}} = \frac{.60}{1.13} = .53$$

Clearly, the coefficient of .30 considerably underestimates the validity of the test.

Estimating validity from incidental selection It might also be the case that we want to estimate what the validity coefficient would be for the full range from knowledge of the correlation in the restricted sample and the restricted and unrestricted variance in the variable subject to *incidental selection*. That is, if the variance in the predictor (X) were restricted directly by accepting only those people above a certain cutoff score, but we were only able to obtain data for the criterion (Y), we wish to estimate r_{xy} given knowledge of r'_{xy} , σ_y , and σ'_y .

The appropriate solution may be obtained by beginning with Equation 10-9.

$$\sigma'_y \sqrt{1 - r'^2_{xy}} = \sigma_y \sqrt{1 - r^2_{xy}}$$

If we simply square both sides, divide by σ_y^2 , and solve for r_{xy} , the result is

$$r_{xy} = \sqrt{1 - (1 - r'^2_{xy}) \frac{\sigma_y'^2}{\sigma_y^2}} \quad (10-13)$$

Most likely, a need for this formula would seldom arise in practice. However, one should be extremely careful not to confuse calculating r_{xy} from knowledge of incidental restriction versus knowledge of direct restriction. Equations 10-12 and 10-13 are not the same. Make sure you match your situation with the appropriate formula.

Another note of caution: It could be the case that *direct* restriction has taken place with regard to the criterion (Y), not the predictor (X). For example, suppose a test is given to all applicants, but is not used in selection. Subsequently 25 percent of the low performers on the criterion (Y) are explicitly asked to leave the organization. In this case the direct restriction is on the criterion (Y), and the role of the predictor and criterion in the previous formulas should be *reversed*.

We might also wish to know how the explicit selection on the predictor affects the *variation* in the criterion rather than the correlation between predictor and criterion. The extent of incidental variation in the criterion is indicated by σ'_y/σ_y . Dividing both sides of Equation 10-8 by σ_y gives

$$\frac{\sigma'_y}{\sigma_y} = \frac{r_{xy}(\sigma'_x/\sigma_x)}{r'_{xy}} \quad (10-14)$$

Substituting for r'_{xy} from Equation 10-11 and simplifying gives

$$\frac{\sigma'_y}{\sigma_y} = 1 - r^2_{xy} + r^2_{xy} \frac{\sigma_x'^2}{\sigma_x^2} \quad (10-15)$$

To show the incidental effects on the variation of criterion scores of the explicit restriction of range on the predictor scores, Equation 10-15 is plotted in Figure 10-2. In this figure we can see that the greater the deliberate restriction on the predictor, the greater the incidental restriction on the criterion.

If we wished to express the incidental restriction on Y in terms of r'_{xy} , the correlation in the restricted sample, rather than r_{xy} , the correlation across the entire range, we should substitute for r_{xy} from Equation 10-12 in Equation 10-15 and simplify. Then

$$\sigma_y = \sigma'_y \left(1 - r'^2_{xy} + r'^2_{xy} \frac{\sigma_x'^2}{\sigma_x^2} \right) \quad (10-16)$$

In the foregoing example of the selection of office personnel, let us say that the standard deviation of the criterion scores of those persons who were hired is 8. Variation in criterion scores has been subjected to incidental restriction. We estimate the standard deviation of criterion scores had all applicants been hired from the above formula as follows:

$$\sigma_y = 8 \left(1 - (.30)^2 + (.30)^2 \frac{(10)^2}{(5)^2} \right) = 8(1.27) = 10.16$$

Restriction of range in three-variable situation Here we are concerned with the deliberate or explicit restriction of range of scores on one variable, Z , on the validity with which X predicts Y . The ratio σ'_z/σ_z describes the amount of explicit restriction of scores on variable Z , the ratio σ'_x/σ_x describes the amount of incidental restriction of predictor scores, and the ratio σ'_y/σ_y describes the amount of incidental restriction on criterion scores.

In addition to the two assumptions we have already made that

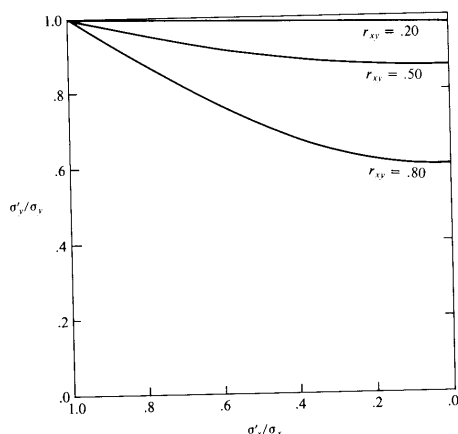


Figure 10-2 Incidental effects of deliberate or explicit restriction of range of predictor (X) scores on variation in criterion (Y) scores.

$b'_{y \cdot x} = b_{y \cdot x}$ and $\sigma'_{y \cdot x} = \sigma_{y \cdot x}$, we must assume that the two partial correlations are the same. That is, we assume

$$r'_{xy \cdot z} = r_{xy \cdot z} \quad (10-17)$$

We are here making the not unreasonable assumption that if we hold scores of variable Z constant, the correlation between X and Y scores will be the same regardless of whether we have only a restricted range of Z scores or the entire range. Since Z is the variable on which there is a deliberate restriction of range, from Equation 10-7 we can write

$$r'_{zx} \frac{\sigma'_x}{\sigma'_z} = r_{zx} \frac{\sigma_x}{\sigma_z} \quad (10-18)$$

and

$$r'_{zy} \frac{\sigma'_y}{\sigma'_z} = r_{zy} \frac{\sigma_y}{\sigma_z} \quad (10-19)$$

Solving these two equations for r'_{zx} and r'_{zy} , we have

$$r'_{zx} = r_{zx} \frac{\sigma_x \sigma'_z}{\sigma'_x \sigma_z} \quad (10-20)$$

and

$$r'_{zy} = r_{zy} \frac{\sigma_y \sigma'_z}{\sigma'_y \sigma_z} \quad (10-21)$$

Since Z is the variable on which there is deliberate restriction of range, we can write

$$\sigma'_x \sqrt{1 - r'^2_{zx}} = \sigma_z \sqrt{1 - r^2_{zx}}$$

and

$$\sigma'_y \sqrt{1 - r'^2_{zy}} = \sigma_z \sqrt{1 - r^2_{zy}}$$

Dividing both sides of the first of these two equations by σ'_x and both sides of the second equation by σ'_y gives us

$$\sqrt{1 - r'^2_{zx}} = \frac{\sigma_z}{\sigma'_x} \sqrt{1 - r^2_{zx}} \quad (10-22)$$

and

$$\sqrt{1 - r'^2_{zy}} = \frac{\sigma_z}{\sigma'_y} \sqrt{1 - r^2_{zy}} \quad (10-23)$$

Since by assumption $r'_{xy \cdot z} = r_{xy \cdot z}$, we can write the equality of these two partial coefficients of correlation as

$$\frac{r'_{xy} - r'_{zx} r'_{zy}}{\sqrt{(1 - r'^2_{zx})(1 - r'^2_{zy})}} = \frac{r_{xy} - r_{zx} r_{zy}}{\sqrt{(1 - r^2_{zx})(1 - r^2_{zy})}}$$

In this equation we can substitute for $\sqrt{1 - r'^2_{zx}}$ from Equation 10-22 and for $\sqrt{1 - r'^2_{zy}}$ from Equation 10-23.

$$\frac{r'_{xy} - r'_{zx} r'_{zy}}{(\sigma'_x / \sigma'_x) \sqrt{1 - r^2_{zx}} (\sigma_y / \sigma'_y) \sqrt{1 - r^2_{zy}}} = \frac{r_{xy} - r_{zx} r_{zy}}{\sqrt{(1 - r^2_{zx})(1 - r^2_{zy})}}$$

Multiplying both sides of the equation by $\sqrt{(1 - r^2_{zx})(1 - r^2_{zy})}$ gives

$$\begin{aligned} \frac{r'_{xy} - r'_{zx} r'_{zy}}{\sigma'_x \sigma'_y / \sigma'_x \sigma'_y} &= r_{xy} - r_{zx} r_{zy} \\ r'_{xy} &= (r_{xy} - r_{zx} r_{zy}) \frac{\sigma_x \sigma_y}{\sigma'_x \sigma'_y} + r'_{zx} r'_{zy} \end{aligned}$$

Substituting for r'_{zx} and r'_{zy} from Equations 10-20 and 10-21 gives

$$\begin{aligned} r'_{xy} &= (r_{xy} - r_{zx} r_{zy}) \frac{\sigma_x \sigma_y}{\sigma'_x \sigma'_y} + r_{zx} \frac{\sigma_x \sigma'_z}{\sigma'_x \sigma_z} r_{zy} \frac{\sigma_y \sigma'_z}{\sigma'_y \sigma_z} \\ &= \frac{\sigma_x \sigma_y}{\sigma'_x \sigma'_y} \left(r_{xy} - r_{zx} r_{zy} + r_{zx} r_{zy} \frac{\sigma_z^2}{\sigma_x^2 \sigma_y^2} \right) \end{aligned} \quad (10-24)$$

Recalling that the explicit restriction is on variable Z and the incidental restriction is on both variables, X and Y , from Equation 10-15 we can write

$$\frac{\sigma'_x}{\sigma_x} = \sqrt{1 - r_{zx}^2 + r_{zx}^2 \frac{\sigma_z'^2}{\sigma_z^2}}$$

and

$$\frac{\sigma'_y}{\sigma_y} = \sqrt{1 - r_{zy}^2 + r_{zy}^2 \frac{\sigma_z'^2}{\sigma_z^2}}$$

Therefore

$$\frac{\sigma_x \sigma_y}{\sigma'_x \sigma'_y} = \frac{1}{\sqrt{[1 - r_{zx}^2 + r_{zx}^2 (\sigma_z'^2 / \sigma_z^2)][1 - r_{zy}^2 + r_{zy}^2 (\sigma_z'^2 / \sigma_z^2)]}} \quad (10-25)$$

Substituting for $\sigma_x \sigma_y / \sigma'_x \sigma'_y$ from Equation 10-25 in Equation 10-24 gives

$$r'_{xy} = \frac{r_{xy} - r_{zx} r_{zy} + r_{zx} r_{zy} (\sigma_z'^2 / \sigma_z^2)}{\sqrt{[1 - r_{zx}^2 + r_{zx}^2 (\sigma_z'^2 / \sigma_z^2)][1 - r_{zy}^2 + r_{zy}^2 (\sigma_z'^2 / \sigma_z^2)]}} \quad (10-26)$$

Equation 10-26 shows what the validity of a predictor, r'_{xy} , would be if we deliberately reduced the variation in scores on variable Z from σ_z to σ'_z . Although not really complicated, the above formulas are long and noxious-appearing, so it may (or may not) help matters to consult Figure 10-3. In this figure we can see that as the variation in Z scores is reduced, that is, as σ'_z / σ_z becomes smaller, the validity of the test, r'_{xy} , also becomes smaller. It will also be seen that the lower the correlation between the restricted variable (Z) and the predictor (X) and the criterion (Y), the less will be the effect of restriction of range on the validity coefficient. In some extreme cases where the correlation between the restricted variable and both the predictor and the criterion is higher than the correlation between the predictor and the criterion, restriction of range may even change a positive relationship to a negative one. It is therefore obvious that if a test is validated on a group whose scores are restricted on another variable, which is related to both the predictor and the criterion, the validity coefficient will be underestimated.

Equation 10-26 shows that the validity coefficient will be reduced as the variation in scores on variable Z is reduced. Ordinarily we have data on a group with restricted range and wish to estimate what the validity coefficient would be if we had the total range. That is, we want to know r_{xy} rather than r'_{xy} . To obtain this value, we start with Equations 10-18 and 10-19, solving for r_{zx} and r_{zy} . The solution is parallel to that for r'_{xy} , so that, equivalent to Equation 10-26, we can write

$$r_{xy} = \frac{r'_{xy} - r'_{zx} r'_{zy} + r'_{zx} r'_{zy} (\sigma_z^2 / \sigma_z'^2)}{\sqrt{[1 - r_{zx}'^2 + r_{zx}'^2 (\sigma_z^2 / \sigma_z'^2)][1 - r_{zy}'^2 + r_{zy}'^2 (\sigma_z^2 / \sigma_z'^2)]}} \quad (10-27)$$

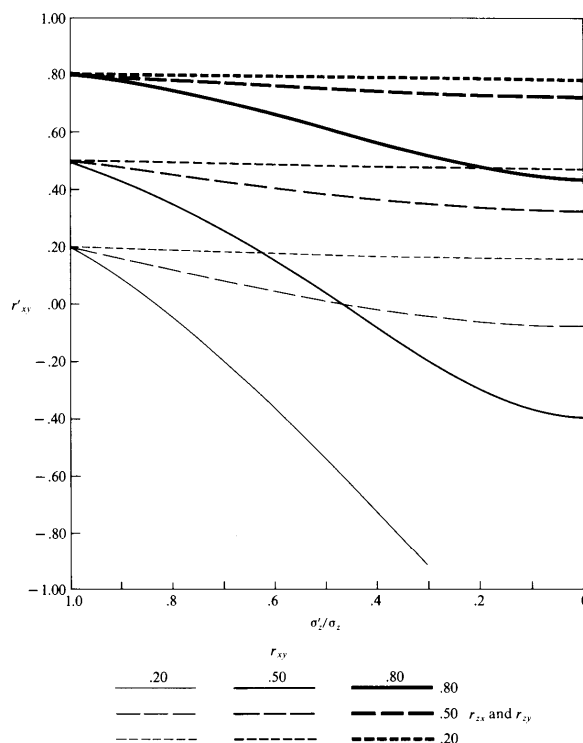


Figure 10-3 Effects of deliberate or explicit restriction of range of scores on variable Z on validity coefficient (r_{xy}).

Suppose that students are admitted to college on the basis of their high school grade average, the standard deviation of the grade average of all applicants being .50 and that for those who are accepted being .25. Let us say that we wish to determine the validity of a scholastic aptitude test and administer it to entering freshmen, the high school graduates who have acceptable grades. Suppose we find with this selected group that the validity of high school grade average, Z , in predicting college grades is .30 (r'_{zy}), the validity of the scholastic aptitude test is .40 (r'_{xy}), and the correlation between high school grade average and scores on

the scholastic aptitude test is .20 (r'_{zx}). We can estimate the validity for the entire range of applicants (r_{xy}) from Equation 10-27 as follows:

$$r_{xy} = \frac{.40 - (.20)(.30) + (.20)(.30)[(.50)^2/ (.25)^2]}{\sqrt{[1 - (.20)^2 + (.20)^2[(.50)^2/ (.25)^2]]\{1 - (.30)^2 + (.30)^2[(.50)^2/ (.25)^2]\}}}$$

$$= \frac{.58}{\sqrt{1.42}} = .49$$

Summary comment on restriction of range Our attempt to keep this story short was not very successful, and we apologize for that. Do not get too bogged down in the details of the formulas. The important things are to understand the assumptions on which they are based and to match the data you have with the parameters you want to estimate. Make sure the distinctions between direct and incidental selection and the variables involved in each are kept clear.

When used appropriately, the correction for reduction of range is a valuable aid in the interpretation of criterion-related validity. It can help give a fairer picture of what the actual validity of a predictor will be.

Alternative Indices of Validity

During recent years considerable discussion has centered around the appropriate index for assessing criterion-related validity. As we have already noted, the conventional index is the Pearsonian correlation coefficient, which is imbedded in the so-called classical validity model in the form of the normal bivariate distribution. The model implies that (1) there is one normally distributed criterion or criterion composite, (2) there is one normally distributed predictor or predictor composite, (3) the relationship between them is linear and homoscedastic, and (4) the index of validity is some form of the product moment correlation coefficient.

Alternatives to the one predictor-one criterion configuration have been proposed, and these are discussed elsewhere in this book. What concerns us here is the role of the correlation coefficient itself.

As an index of the level of practical predictive accuracy, the product moment coefficient is not that easy to translate into meaningful terms. How useful is a value of .55? Most people have a feeling that .55 is reasonably good, but they would be hard-pressed to say why explicitly. Pointing to the relative magnitude of the standard error of estimate [$SE_{est} = \sigma_y \sqrt{1 - r_{xy}^2}$] is one kind of answer, but even here the meaning is not always perfectly clear.

Decision Accuracy

Another alternative is to think of the prediction problem as a task in predicting *discrete* criterion outcomes. Almost all practical situations can be translated into the problem of predicting "criterion categories" of various kinds, the most frequent being success and failure. Once the problem is recast in these terms, it becomes meaningful to talk about validity, or accuracy, as the proportion of correct predictions.

If the variable to be predicted is a dichotomy, the bivariate distribution reduces to a fourfold table and four different outcomes are possible: (1) correctly predict the high-criterion category, (2) correctly predict the low-criterion category, (3) overpredict (false positive), and (4) underpredict (false negative). This simple arrangement is shown in Figure 10-4.

Certain things can be discerned from looking at the figure. In general, the relative proportion of correct predictions versus errors in prediction is a function of three principal factors. The first factor is the criterion "split" that defines the two criterion categories (for example, success/failure). Although this statement is offered without proof, it is true that (other things being equal) the greatest *gains* in accuracy from using new and more valid predictors occur when the split is 50-50. In general, this is because the variance of a dichotomous variable is equal to $P \times Q$, where P and Q are the two proportions, and the variance is at a maximum when $P = Q = .50$. As we have seen before, the greater the variance, the greater the potential relationship with the predictor. The proportion of people in the high-criterion group is often called the *base rate for success*, or simply the *base rate*.

The second factor is the *cutting score* on the predictor, or the predictor score above which we predict success and below which we predict failure. The cutting score is under the control of the decisionmaker, and one can see that as it is set higher or lower, the relative number of observations in the four cells will change accordingly. However, by changing the cutting score, and nothing else, we can reduce the frequency of one kind of error only at the expense of increasing the second kind.

Finally, the relative number of hits and misses is a function of the degree of association between the predictor and the criterion. The higher the correlation, the "narrower" the bivariate distribution; and the greater the percentage of cases in cells *A* and *B*, the fewer the percentage of cases in cells *C* and *D*.

One can also see from the figure that the two different kinds of errors are affected differentially by changes in the cutting score and in the base rate. For example, moving the cutting score to the right reduces the proportion of false positives at the expense of more false negatives.

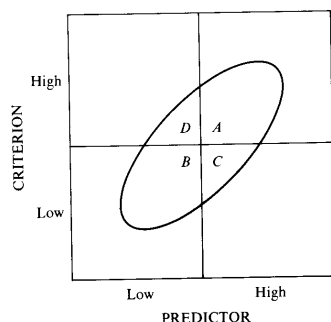


Figure 10-4 Continuous bivariate distribution between predictor and criterion reduced to dichotomous prediction decision.

The Taylor-Russell tables (1939), shown in Appendix B, take advantage of this fact and portray the increase in the percentage of correct predictions over and above the base rate as the cutting score is raised, given a particular *selection ratio* (percentage hired) and correlation coefficient. The tables do this by assuming that both predictor and criterion are actually continuous and that their joint distribution is bivariate normal. Then the mathematical description of the bivariate distribution can be used to specify the areas (percentage of cases) above or below any particular cutting score or base rate. However, the tables deal with only one kind of error—false positives—and one kind of correct prediction—high hits. Thus, in the Taylor-Russell sense the increase in the percentage of correct predictions is

$$\% \text{ increase} = \frac{A}{A + C} - \frac{A + D}{A + B + C + D}$$

The latter quantity is simply the base rate. What is often forgotten is that this choice of an accuracy statistic is very much a value judgment by somebody. Traditionally, the false positive (cell C in Figure 10-4) may indeed have been the prediction mistake to avoid in personnel or educational selection. However, the weight of public policy is often in favor of minimizing the false negative (cell D), especially with regard to minority and disadvantaged groups. It seems safe to predict that our society will endure considerable conflict and soul-searching before the issue of who bears the implicit and explicit costs of both false positives and false negatives is decided. Nevertheless, it must be considered, and

the issue is obscured by looking only at the correlation coefficient as an index of accuracy.

Relative to the base rate, maximum gain in decision accuracy from a specific validity coefficient is obtained when the base rate equals .50. Again this is nothing more than a restatement of the fact that for a dichotomous variable, the variance is at a maximum when $P = Q = .50$. As the base rate becomes more extreme in either direction, achieving large increases in the number of correct decisions becomes progressively more difficult (that is, it takes greater validity and/or smaller selection ratios). At some point, when the increase in correct decisions is compared to the cost of prediction, the prediction system may not seem worth it. If the proportion of people who fail is very small, it may make more sense to use the base rate and predict "success" for everyone. In a selection situation we might then select on the basis of first-come/first-served, random assignment, or some other strategy deemed as equitable by prevailing community standards. If the proportion of people who succeed is very small, the situation is a bit different. In a personnel selection situation nobody would be hired if we used the base rate to predict. It would make more sense to restructure the applicant pool to increase the relative number of talented people and thus make the base rate less extreme. To take an obvious but extreme example, we could accept as applicants only those individuals who have successfully performed the job in the past in some other organization (for example, university presidents).

As will be explained more fully in a moment, we can assert that if the predictor has greater than zero validity, there will almost always be *some* improvement in predictive accuracy in excess of the base rate, although it may not be great. The "almost" will be examined below. It is also true that under certain conditions the base rate can be beaten even when a predictor has *zero* validity (in the correlational sense), but let's ignore these unnamed conditions for the moment.

If, as in the Taylor-Russell tables, the primary concern is with the false positives, the base rate can be beaten by setting a higher and higher predictor cutting score. However, if the total number of errors ($C + D$) is important, then the base rate can still be beaten if the cutting score is set at the point where the predictor distributions for the high- and low-criterion groups intersect as in Figure 10-5a. This rule holds even for peculiar distributions, as when there are two points of intersection (Figure 10-5b). In this instance, applicants would be accepted if they scored between the two cutting scores (CS_1 and CS_2) and rejected otherwise. The classic example concerns aptitude measures as a predictor and a criterion that individuals below CS_1 are not capable of meeting and people above CS_2 are too uninterested to meet. However, we should keep in mind that if the cutting scores are not set properly,

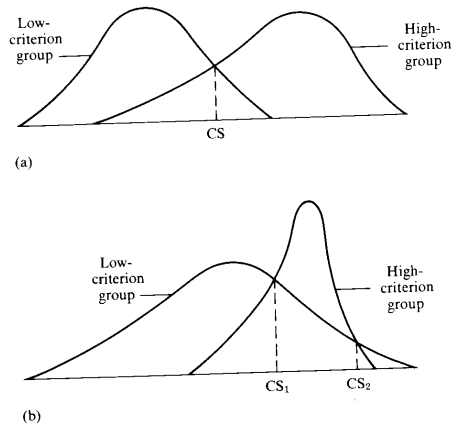


Figure 10-5 Optimal cutting scores when objective is to minimize total number of prediction errors (that is, false positives + false negatives).

even a valid predictor can yield less accuracy than the base rate. For example, if the base rate is high (a large percentage of the unselected group would succeed) and the cutting score is set too high, a slightly higher percentage of high hits will be identified at the cost of many more false negatives.

Notice that focusing on the specific proportion of correct predictions in a particular situation, rather than the correlation coefficient, makes no assumptions about the form of the distribution and takes advantage of the nonlinear as well as linear components of association between the two variables. If the bivariate distribution is heteroscedastic, then predictions may be better at one end than the other (that is, successes or failures). As always, we are faced with the question of whether departures from the normal bivariate distribution are real or merely reflect sampling error. If we opt for the latter, then it might be more advantageous in the long run to "smooth out" the frequencies in the fourfold table by computing the product moment correlation and converting the coefficient to the corresponding proportions of hits and misses via the Taylor-Russell tables, if both predictor and criterion are dichotomous. However, even though the correlation coefficient may underestimate the degree of association in a truly curvilinear relationship, if the variables are continuous, the phi coefficient will overestimate the relative

number of correct predictions when the variables are dichotomous and the splits depart very much from 50-50. Thus under these conditions entering a phi coefficient in the Taylor-Russell tables would be inappropriate.

At this point it might be informative to ask how the correlation coefficient is related to the proportion of correct predictions if the bivariate distribution is indeed linear and homoscedastic. Curtis and Alf (1969) have shown that the proportion of total correct predictions—that is, considering both kinds $(A + B)/(A + B + C + D)$ —is a near-linear function of r , not r^2 . Thus an increase in predictability from $r_{xy} = .20$ to $r_{xy} = .40$ is as valuable as an increase from .50 to .70, providing we accept the proportion of total correct predictions as the appropriate index. Further, Brogden (1946) has shown that the difference, or increase, in the criterion mean when the selected group is compared to the unselected group is also a linear function of r_{xy} .

A more elaborate set of tables for converting a correlation coefficient to frequencies of correct predictions has been provided by Tiffin and McCormick (1965). In the Tiffin and McCormick tables the predictor distribution (assumed to be normal) is divided into five equal portions (0-20th percentile, 20-40th percentile, and so on). For a specific criterion base rate and a specific value for the validity coefficient, the percentage of people in each score range on the predictor who fall in the high-criterion, or success, category is specified. The situation is as portrayed in Figure 10-6.

The tables presented by Tiffin and McCormick are called *expectancy tables* or *expectancy charts*. When the percentages of expected successes are obtained by using the areas specified by the normal bivariate distribution corresponding to a particular correlation, the table is called a *theoretical expectancy chart*. If the observed percentages obtained from a specific sample are entered in the table, then it is referred to as an *empirical expectancy chart*.

The correlation coefficient can be transformed to its equivalent increase in the criterion mean via the Naylor-Shine tables (Blum and Naylor, 1968). These are analogous to the Taylor-Russell tables, but the focus is on the increase in the mean criterion score, not the increase in the percentage of correct predictions. The existing criterion distribution in the organization is assumed to be unit normal, and the validity coefficient is taken as a concurrent validity coefficient computed on the existing organizational population that has been selected by whatever procedures were previously in use. Then for a particular selection ratio, the increase in the criterion mean (in standard-score units) that would be obtained by employing the new selection method can be read from the table. Keep in mind that using the Naylor-Shine tables, or for that matter the Taylor-Russell tables, to convert r_{xy} to a more meaningful index is dependent on the degree of linear association.

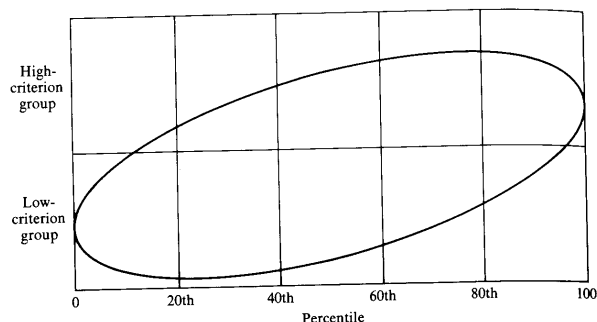


Figure 10-6 Normal bivariate distribution partitioned by dichotomous criterion and predictor distribution divided into quintiles.

More Complex Situations

The above discussion of cutting scores and errors of prediction has been fairly simpleminded in terms of all the relevant parameters that might possibly be involved. Again the appropriate portrayal of the cutting score(s) and the resulting prediction errors is a plot of the predictor distributions for both criterion groups, as in Figure 10-5. Consider a list of all the parameters contained in this picture:

1. The difference between the means of the two distributions (validity).
2. The relative number of observations in the two distributions (base rate).
3. The cutting score(s).
4. The shape of each distribution.
5. The difference between the standard deviations of each distribution.

So far we have discussed some of the implications of differences in validity, base rate, and cutting score, as well as touching on the problem of the shape of the distribution. We have said nothing about what happens when the variability of the two distributions is different and have been implicitly assuming they are the same.

We observed previously that if the base rates depart very far from 50-50, relatively small changes in the cutting score can lead to relatively large changes in the number of prediction errors. The same

danger arises when the standard deviations of the two distributions begin to depart from each other. When the base rate departs from 50-50 and the standard deviations depart from equality, things really begin to get complicated. For example, it is possible for no cutting scores to exist (as in Figure 10-7a). Here one distribution is completely contained in the other because the base rate for success is very high, meaning that not very many people "fail," and the standard deviation for the low-criterion group is relatively small. In this situation, *even though the mean predictor scores for the two groups differ*, any cutting score that is chosen will yield many more additional errors than additional correct predictions.

On the other side of the coin, it is also true that even though the predictor means for the two criterion groups may be identical (as in Figure 10-7b), and "validity" therefore is zero, the proper cutting scores may still be able to improve the accuracy of prediction over and above the base rate. The leverage for this seemingly incongruous result is provided by the difference in standard deviations. One can "fail" on this criterion by possessing too much or too little of the attribute measured by the predictor score (intelligence, interest in artistic/creative tasks, and so on). It is the people who possess a moderate amount that do best. Consequently, by setting two cutting scores and predicting success for those in the middle, the increase in correct predictions is greater than the increase in errors of prediction.

From Accuracy to Utility

We have now moved from a notion of predictive validity in the form of a correlation coefficient to predictive accuracy in the form of the proportion of correct predictions in excess of the base rate. One could still wonder about how to interpret the value of a particular increase in predictive accuracy. Given a specific base rate, is the value of a 15 percent increase in predictive accuracy always the same?

In their now classic statements on this issue, Brogden (1949) and Cronbach and Gleser (1965) have reminded us that it is not. The *value* of an increase in predictive accuracy is dependent on the *costs* associated with obtaining it versus the *payoffs* or benefits obtained from each additional correct prediction. For example, in most situations the costs of increasing predictive accuracy for selecting people into managerial positions may be considerable, but they pale beside the payoff from making even one more correct selection decision. In short this argument moves us from a concern with predictive validity and predictive accuracy to a concern for predictive usefulness.

The Cronbach and Gleser presentation of these ideas has become a

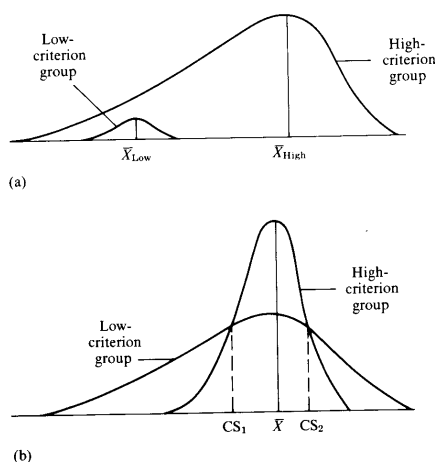


Figure 10-7 Effects of extreme differences in base rates and predictor variances on dichotomous prediction decisions. (a) Predict success for everyone. (b) Predict success for individuals between CS_1 and CS_2 .

classic in spite of the fact that their prescriptions for how to judge utility make considerable demands on our measurement technology and may not always be possible to implement. Much of the value of these formulations lies instead in posing the very central issues that always must be considered in any prediction situation, however imperfectly, and they deserve at least a brief summary here.

The basic notions were first advanced by Brogden (1949) and went something like this. Suppose criterion scores (Y) are expressed in terms of the overall value or benefit they represent (for example, an individual performer's economic value, in dollar terms, to the organization). Further recall that the simple linear regression coefficient $b_{y \cdot x}$ discussed in Chapters 5 and 6 can be represented as

$$b_{y \cdot x} = r_{xy} \left(\frac{\sigma_y}{\sigma_x} \right)$$

As before, the simple linear regression equation then becomes

$$Y'_i = A_{y \cdot x} + r_{xy} \left(\frac{\sigma_y}{\sigma_x} \right) \cdot X_i$$

Now suppose we express the predictor (X) in standard scores and the criterion (Y) in deviation scores. The regression equation then becomes

$$Y'_i - \bar{Y} = r_{xy}(\sigma_y)(z_{x_i})$$

The constant A disappears because we are subtracting the mean of the criterion scores from each individual score, and thus the intercept (A) moves to zero on the y axis. The standard deviation of the predictor score disappears because converting to standard scores makes $\sigma_x = 1.0$.

If the validity coefficient r_{xy} is positive (hopefully), then $Y'_i - \bar{Y}$ is the increase in utility to be expected from using the predictor, as compared to random selection; that is, predicting $\bar{Y}$ for everybody. This quantity is usually referred to as Δu_i (for the change in utility obtained by using the predictor).

To go just one step further, what we might like to have is the average gain in utility per person. One thing we could do is compute Δu_i for each person and average them. However, if we are willing to assume that the predictor is normally distributed (still with mean zero and $\sigma_x = 1.0$), then the mean predictor score for those selected is $E(x')$, which is the height (or ordinate) of the normal curve at the point on the x axis corresponding to the selection ratio. The equation for the average utility gain then becomes

$$\Delta u = r_{xy} \sigma_y E(x') \quad (10-28)$$

It remained for Cronbach and Gleser (1965) to remind everyone that few things in life are free, and besides the benefits to be gained from using a particular predictor, there are also costs to be considered. Costs refer to such things as buying test booklets, training and/or paying additional interviewers, and the like. For some situations costs may be low (administering a 15-minute test), and in some cases they may be high (multiple interviews by a number of people flown to a central location).

Given the added component of cost, Cronbach and Gleser's basic definitional formula for the average gain in value associated with a predictor

$$\Delta u = r_{xy} \sigma_y E(x') - C_x \quad (10-29)$$

where

Δu = net gain in utility

r_{xy} = validity coefficient obtained concurrently on present organizational population

σ_y = standard deviation of payoff (criterion) distribution

$E(x')$ = ordinate of unit normal distribution at point on base line corresponding to cutting score (*selection ratio*) on predictor distribution

C_x = cost of obtaining predictor information

To use this algorithm effectively, it should be computed for a variety of alternative predictors, including the one currently in use. If this could be done, the interaction of the validity and costs of prediction for different predictors would be made quite clear.

The problem, of course, is the metric for the payoff distribution. It must reflect the differential benefits of correct predictions associated with different jobs, different organizational needs, and so on. It seems obvious that the value for σ_y would vary tremendously across situations. Also since the basic algorithm involves the correlation coefficient, another problem is the implication that payoff is linearly associated with the predictor continuum and that false positives and false negatives are given equal weight.

A number of interpretive principles flow from the utility-theory view, and Cronbach and Gleser worked out the mathematics for many of them. We summarize only a few of them here:

1. The predictor with the highest validity coefficient may not have the highest utility, compared to alternatives, if
 - a. It is proportionately more costly than it is more valid.
 - b. It predicts a criterion that is proportionately less important.
2. Even predictors with low validities may be very useful if σ_y is large and the selection ratio is not too extreme.
3. If "adaptive treatments" are possible, then the gain in utility, given a particular gain in validity, is no longer a linear function, but becomes positively accelerated. For example, a validity increase of .40 to .60 is worth more and more to the extent that the sample can be partitioned, on the basis of the predictor score, and different training or different task assignments provided for the people in each category. For example, using a math aptitude test to select people for math instruction is more and more valuable to the extent that, high, low, and medium scorers can be given instruction appropriate for their level of readiness.
4. Relative to the choice among using two tests as a battery (that is, weighted sum), sequentially (for example, successive hurdles), or singly (just one of them), the model replies:
 - a. As the selection ratio moves from .50 to the extremes, the strategy of choice goes from using both tests as a battery to using them sequentially, to using just one test. This pro-

gression is speeded up to the extent that the tests are intercorrelated and have different costs associated with them. Obviously, if the costs are quite different and the predictors are highly intercorrelated, it literally "pays" to use the cheapest one.

- b. A similar progression of choices exists as the difference between the two validity coefficients becomes greater and greater—other things, such as intercorrelations and costs, being equal.
5. In sequential testing, the predictor given first is the one with the greatest difference between cost and benefit. This is in contrast to the more conventional wisdom, which says to use the most "valid" predictor first.

In sum, the Cronbach and Gleser treatment highlights a number of considerations that people interested in practical decisionmaking should keep in mind, although perfect answers to the questions posed by the model aren't always possible.

Further Utility Considerations

It may indeed be a difficult task to measure utility, but it is there nevertheless; and no matter what type of validity model is used, utilities play an important role even if unrecognized. For example, if we act as if the values of all kinds of errors are equal, we have made an implicit value judgment about utilities that may be so far from the "truth" (as judged by "reasonable" people) that the organization is seriously hurt because of it. If we ignore costs for the time being and again adopt a decisionmaking framework (for example, prediction of a dichotomous outcome), much ground can be regained if the *ratio* of the utilities for correct predictions of success and correct predictions of failure can be reasonably estimated.

Keep firmly in mind here that relative utilities and/or their ratios are value judgments by the management, stockholders, policymakers, citizens' groups, or whoever else has, or should have, a stake in the decision being made. Relative utilities result from whatever policymaking process is operating in a particular situation, and they should be argued about within this framework.

In some prediction situations the cutting scores may be a "given," as when the predictor is a single-test item that is scored right or wrong. For this situation there is a relatively simple method for trying out different utility ratios, such that the relative utility of using the predictor versus using the base rate can be determined.

Consider then the situation portrayed in Figure 10-8. The first matrix

Success	.16	.24	.40
Failure	.48	.12	.60
	Predict failure	Predict success	

(a)

Success	-5	10
Failure	10	-30
	Predict failure	Predict success

(b)

Success	-.80	2.40
Failure	4.80	-3.60
	Predict failure	Predict success

(c)

Figure 10-8 Relative utilities applied to dichotomous prediction decision with fixed cutting score (that is, proportion above CS = .36). (a) Proportion of total for each outcome state. (b) Relative utilities of each outcome. (c) Products of relative frequencies and relative utilities.

shows the proportions of people at each possible outcome. The second matrix shows a hypothetical set of relative utilities for each outcome. The third matrix contains the cross products for each cell.

The relative utility of using the predictor is the sum of the cell entries in matrix c, or +2.8. If we do not use the predictor, we can either predict success for everybody or predict failure for everybody. Our estimate of the base rate is that 40 percent of the population will succeed and 60 percent will fail. If we predict failure for everybody, the relative utility is $(.60 \times 10) + (.40 \times -5) = 4.0$. If we predict success for everybody, the relative utility is $(.60 \times -30) + (.40 \times 10) = -14.0$. Obviously predicting success for everybody is not the thing to do, and predicting failure for everybody is somewhat better than using the predictor. However, if the costs of not filling the job outweigh the gain in overall utility from hiring no one, the predictor should be used. The thing that produced this result is the relatively small utility assigned to someone who succeeds. However, if these relative utilities are deemed reasonable, the moral is that a rough but reasonable assignment of relative utilities can give a more detailed picture of what's going on in a selection situation than we have without an assignment of relative utilities.

Summary Comment on Criterion-Related Indices

There is no easy "cookbook" one can follow to select the way in which criterion-related validity should be portrayed. If the linear and homoscedastic correlation model is deemed an appropriate representation of the form of the relationship between predictor and criterion, then the correlation coefficient is the place to start. It is an indication of the extent to which the variability in the criterion can be accounted for by the factors measured by the predictor. For certain conceptual ques-

tions (for example, "To what extent does garden-variety anxiety relate to number of days of illness per year?") that may be sufficient and appropriate.

However, criterion-related validities are usually obtained to aid us in making some kind of decision about individuals (hire/not hire, admit/not admit, treat/not treat) and therefore should be interpreted in a decisionmaking context. When this is done, we must face questions such as: How many predictor (decision) categories should there be? How many criterion (outcome) categories should there be? How should they be defined; that is, what are the base rate and cutting score(s)? When viewed this way, it is readily apparent that a given validity (correlation) coefficient may be more or less useful depending on the base rate and cutting score (selection ratio). For example, if the base rate is near 50-50 and the cutting score can be set fairly high, even very small validity coefficients can improve decisionmaking accuracy by a significant amount. In general the crucial consideration here is the precise specification of the decision for which the information is to be used.

After the nature of the decision under consideration has been specified, questions of costs and benefits must also be considered. Even if costs and benefits can't be measured precisely, "all available" information should be examined and the value judgments of the relevant people involved should be reviewed. To use a predictor wisely, we simply must find out as much as possible about the costs of prediction (both direct and indirect), the benefits to be gained from correct predictions, and the costs of making errors. If there are genuine conflicts over the magnitude of the consequences of various errors, these must be confronted directly. For example, is it worse to admit people who will fail than not to admit someone who would have succeeded? Psychometric theory cannot provide the answers. Society itself must do so.

Equal Opportunity and Concepts of Validity and Fairness

There is one last complicating factor in the criterion-related validity picture that we must discuss. While it is not really a formal part of psychometric theory, it often becomes the paramount issue when validity models are used in applied settings, and it reflects one of the most fundamental social problems in U.S. society. We are speaking of equal educational and employment opportunity considerations as embodied in Title VII of the Civil Rights Act of 1964. Does a predictor discriminate against members of various minority groups in employment or educational selection? Can a predictor be valid and discriminatory (un-

fair) at the same time? If so, what can be done to insure that prediction decisions are both valid *and* fair?

These are important questions, and they are most often considered in the context of comparing the effects of prediction rules for women versus men or minority versus majority groups in educational or job selection. Unfortunately they are not precisely answerable, for reasons we will come to. Also they incorporate a number of important value judgments that cannot be made by appealing to psychometric theory or to some other algorithmic solution. Our brief discussion attempts only to highlight the major issues and to identify some of the more important value judgments.

The situation is further complicated because there are different definitions of discrimination, or fairness, among which it is not easy to choose. In general fairness refers to whether a difference in mean predictor scores between two groups represents a useful distinction for society, relative to a decision that must be made, or whether the difference represents a bias that is irrelevant to the objectives at hand (for example, selecting good people for law school).

When Predictor "Fairness" Is a Problem

In general, in an applied prediction setting, the signal for when to worry about the fairness of a predictor is when there is *adverse impact*. Adverse impact occurs when, for the groups under consideration, the proportion of people selected from each group differs substantially from their relative proportion in the total available applicant population. For example, if a minority group constitutes 35 percent of the available applicants, but only 5 percent of those eventually selected, adverse impact has occurred. The question is then whether the adverse impact is in some sense "justifiable." If the situation involves using one predictor score to make selection decisions, adverse impact is mirrored by a difference in mean predictor scores for the two groups.

How should this difference be interpreted? Does it reflect a corresponding criterion difference? Does it reflect discriminatory practices that have nothing to do with criterion performance? Should different prediction rules be used for the two groups so as to remedy the adverse impact brought on by the group differences in predictor scores? As always, we cannot proceed very far in answering these questions unless we adopt some sort of model of the situation that permits a distinction between validity and fairness. Unfortunately, as you have probably also guessed, there are several models from which to choose. They differ in the assumptions they make and the conclusions (that is, remedies) they suggest. Most of the distinctions are a function of value

judgments that someone must make, and not the result of formal measurement-theory considerations. For this reason the multiplicity of models is useful. That is, they force us to think through what is algebra and what are value judgments, and they help to spell out the specific value judgments that must be made.

Alternative Models of Predictor Fairness

Currently there are a number of models of predictor fairness over which there is much argument, and a full treatment of their algebraic, legal, and national policy implications are far beyond the scope of this book. We can only outline what we think are the most important issues, with due apologies to those people who have contributed much more.

The basic situation considered by almost all models is when the people in each of two groups (male versus female or majority versus minority) are all measured on a predictor X and a criterion Y . Thus the overall univariate distributions on X and Y can be separated into distributions on X and on Y for each subgroup, and the overall bivariate distribution can be separated into bivariate distributions for each subgroup. Consequently there are a mean and a variance for each group on X and Y and a regression line for each group. To simplify things, let us worry no further about differences in variances between the two groups or about whether the two regression lines have unequal slopes. The crucial features are the differences in means for X and Y and the differences in intercepts for the regression lines. Differences in means and intercepts allow for considerable variation in situations that might be observed. Consider the four variations shown in Figure 10-9.

In Figure 10-9a the two groups are on the same regression line (that is, they have the same intercept), and, relatively speaking, the difference between the predictor means is greater than the difference between the criterion means. For the situation Figure 10-9b there is no mean difference in criterion performance, but there is a large mean difference on the predictor. Many people would regard this as a case of pure discrimination. The two groups exhibit comparable criterion performance; but for any cutting score that is set, many more members of group II will be selected. In Figure 10-9c the situation is the converse; while there are no mean differences on the predictor, there is a large difference in mean performance on the criterion. Such a picture may result from discrimination or bias in the performance measure, as when members of a minority group are rated lower (or higher) simply because of their group membership. Although unusual, it is also possible for such a situation to exist in the real world. If groups I and II are female and male respectively, if the predictor is a test of general intelligence,

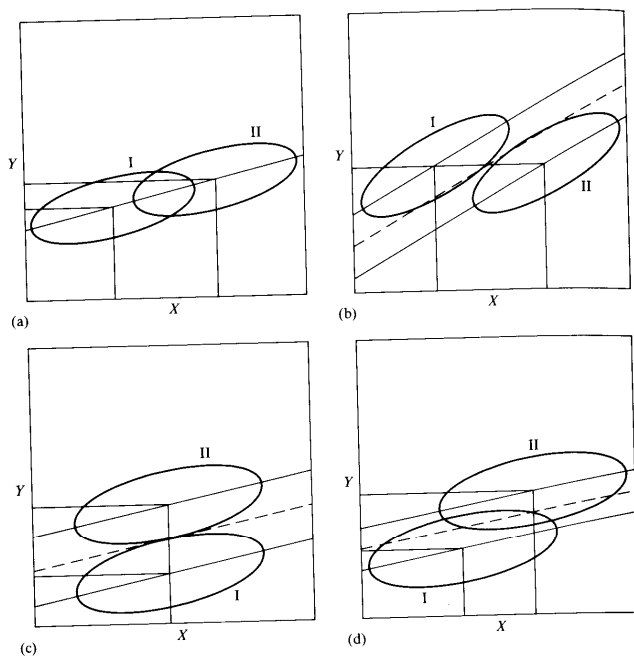


Figure 10-9 Four different bivariate situations when two groups (I and II) are measured on predictor (X) and criterion (Y).

and if the criterion requires *both* intellectual ability *and* physical strength, a picture such as Figure 10-9c could be obtained.

Given this basic portrayal of the validity-and-fairness picture, we can turn to some alternative models for identifying which is which. The thing to watch for is how the models treat the relative differences between the predictor and criterion means for the two groups. Are they "real" differences, or are they the result of discrimination? If it is the latter, what sort of remedy does the model suggest?

Quota model The simplest formulation, and perhaps the easiest to deal with in a public-policy sense, is the *quota model*. From this point of view the difference, if any, in criterion means is ignored, and the difference between predictor means is regarded as bias. The remedy then is

to select people in equal proportions by ranking the people within each group on the predictor and admitting the appropriate percentage of high scorers.

In many situations this might be quite justifiable. It may even be justifiable to go further and select a higher proportion of people from the minority group. The payoff, in terms of providing new role models, for example, may surpass the cost of any loss in validity.

One factor that complicates the quota model is that in real life the two groups of applicants are seldom sampled from comparable populations. That is, one group may be systematically more highly trained or experienced than the other, one group may be older, or in some sense or other one group may be a more highly select sample from its own reference population. Many such differences can result because the recruiting process for the two groups is very different. For example, if professional schools such as law and medicine were guilty in the past of discrimination against women, and admitted only a very few, then it is reasonable to assume that the female applicant pool was much more highly self-selected and therefore of higher average ability than was the male applicant pool. In general, even though the average predictor scores for the two groups may be equal in the general population, the recruitment process seldom draws comparable samples from the population and thus the predictor means may differ somewhat, not because of predictor bias but because of noncomparable recruitment.

One way to account for such nondiscriminatory group differences on the predictor is to examine group differences, if any, on the criterion, in light of the relationship between predictor and criterion (that is, validity). A number of models attempt to specify how these predictor and criterion differences should be interpreted, and it is to some of these that we now turn.

Regression model Within the context of psychometric theory, perhaps the most popular model for conceptualizing predictor fairness is the *regression model* or *classical model* of predictor "bias." This model stipulates that no bias exists if the bivariate distributions for the two groups fall on a common regression line, as in Figure 10-9a. The test for fairness then is to compute the common regression equation for the two groups combined and to determine whether the criterion performance of the people in either group is systematically over- or underpredicted. For example, consider Figure 10-9d in which the dotted line represents the common regression line. For any given predictor score the performance of people in group I will be systematically overpredicted, and the performance of people in group II will be systematically underpredicted. Also, for any given cutting score on X, a greater proportion of group II will be selected.

The remedy suggested by the model is to generate a separate regression equation for each group. That is, the slope $b_{y \cdot x}$ and the intercept $A_{y \cdot x}$ are computed for each group, and each individual's predicted criterion score Y' is computed using the regression line of the appropriate group. The people with the highest predicted criterion scores are then selected. That is, the cutting score is not on the predictor distribution, but on the combined distribution of predicted criterion scores. Such a model will maximize the performance of the selected group and for this reason should be valued by the organization doing the selecting.

The effect of the regression model's remedy on the relative proportions of people selected from the two groups depends on the nature of the situation. In Figure 10-9b using a common regression line would select a greater percentage of group II than group I. The regression model would select them in equal proportions. In Figure 10-9c a single cutting score based on a common regression line would select in equal proportions, but the regression model would take a much higher percentage of people from group II. These are the two extremes. For situations "in between," such as in Figure 10-9d, the relative proportions admitted using the common regression line versus using the regression model must be determined for each specific case.

Regardless of the relative proportion selected in a given situation, it is important to note that the regression model equates fairness to the individual with unbiased prediction. That is, if a prediction rule does not systematically over- or underpredict for individuals with a particular predictor score, it is said to be "fair." Thus the regression model uses the observed group differences in criterion scores to "interpret" group differences on the predictor via the least-squares rule. If, as in Figure 10-9b, there are no criterion differences, then all of the differences between groups on the predictor are interpreted as bias and it is remedied by using separate regression equations.

Equal-risk model The *equal-risk model* of fairness attributed to Einhorn and Bass (1971) is built on a definition of selection fairness proposed by Guion (1966), which states that people with an equal probability of being successful on the job should have an equal probability of being hired. In essence what the equal-risk model does is to argue that selection is fair if the probability of failure (or success) for those people selected is the same for the two groups. To achieve this goal, the strategy is to set the cutting scores for the two groups on the predictor so as to equate the risk. The method begins by selecting a point on the criterion distribution as the minimal acceptable level of performance. Next it considers the people from both groups all of whom have the same predicted criterion score. If the standard error of estimate (that is, the variation within columns) is not the same for the two groups, then the risk (that is, the probability) of falling below the minimal acceptable

level of performance is also not the same for the two groups. The level of risk (proportion of people in the column distribution falling below a certain point on Y) can be adjusted by selecting a different cutting score on X and moving up or down to a different column. Specifying the complete procedure by which the cutting scores are set is a bit too long a story to be recounted here, but the outcome is to select individuals from the two groups who have an equal risk of failure or an equal probability of success.

Thus instead of selecting only people with the highest predicted criterion scores, as does the regression model, the equal-risk model equates the probability of making the success/fail prediction correctly for the two groups. If the regression lines for the two groups are parallel and the standard errors of measurement are equal for the two groups then the equal-risk model is equivalent to the regression model.

Constant-ratio model An example of a different class of fairness models is the *constant-ratio model* originally proposed by Thorndike (1971). This model first partitions the bivariate distribution into the same quadrants, or cells, shown in Figure 10-4 and defines them in the same way. Fairness is then defined as a situation in which, for a particular group, the percentage of people that succeed on the criterion is the same as the percentage that pass the cutting score on the predictor. Thus if 50 percent of a minority group and 70 percent of a majority group succeed on a criterion, but 30 percent of the minority group and 80 percent of the majority group pass the original cutting score on the predictor, two new cutting scores must be set on the predictor such that 50 percent of the minority group and 70 percent of the majority group are selected.

Several things are noteworthy about this model. First it does not deal with continuous criterion scores, but only with the dichotomous success/fail outcome. However, for purposes of specifying which part of the predictor difference is due to discrimination and which is not, it transforms predictor and criterion scores to percentiles. Percentile differences between the criterion means of the two groups are assumed to represent real differences in criterion performance, not bias. Comparing the constant-ratio method to the regression model and the equal-risk model is difficult. They are all the same if the regression slopes and standard errors of measurement are equal and if the mean criterion score is the same for the two groups, as in Figure 10-9b. If the scores of the minority group are generally lower than the scores of the majority group, then the constant-ratio model will usually select a greater proportion of minority applicants than will the other models. In essence, this is because all people who exceed the criterion score that defines success are regarded as equally successful regardless of how high their criterion score actually is. So long as two individuals fall in the success category, the model does not distinguish between one who barely qual-

ifies and one whose performance is two standard deviations higher. This gives the people in the lower distribution a greater chance of being selected than if the actual magnitude of the criterion scores were taken into account.

As regards criterion performance, the central issue here is: "How much is enough?" Can the organization sacrifice some of the top scorers for the sake of selecting a greater number of minority individuals? The answer to this question is the answer to the question of which model is the most appropriate in a given situation.

Summary Comment on Validity and Fairness

With the exception of the quota model, all the fairness models just discussed interpret the mean difference in criterion scores between two groups as a real difference in performance. If there is racial or sex bias in the criterion, then the regression, equal-risk, and constant-ratio models must be further modified or discarded. At this point we can only say that one should worry about this issue and should try by every means possible to develop criterion measures that are free of such bias.

Assuming that criterion differences do, however, represent real performance differences, how then should we choose among the available models for promoting fairness? Ideally, two things are needed. First, we must have some notion of what the payoff (criterion) distribution, in Cronbach and Gleser terms, looks like. That is, what are higher and higher criterion scores worth to the organization? Does payoff become greater and greater, or do higher performance scores not produce much additional payoff once performance exceeds a certain critical point? Second, value judgments must be made as to the utility of correctly selecting additional members of minority groups, failing to select minority-group members who would have been successful, failing to select majority-group members who would be successful, and so on. Once these value judgments are made, it will be possible to choose among fairness models. We would then have some basis for trading off lower (or higher) average criterion scores for the increased (or decreased) participation of specific groups. These initial and fundamental questions are questions of value, and we all must face them.

Summary

Perhaps the most fundamental distinction we can make within the entire topic of validity is that there are two principal themes. One theme

concerns the validity of a predictor (or set of predictors) for making a specific prediction or decision. As we argued at some length in this chapter, the variable of principal interest here is the *criterion*. The predictor is of interest only insofar as it can be used as a substitute for the criterion. In this context we discussed a number of specific *criterion-related validity situations* (for example, predictive, concurrent, postdictive) and the identifying characteristics of each.

Keep in mind that while the objective of criterion-centered validity is to identify predictors that can be used in place of the criterion because the latter is too expensive or not available, at least one sample of individuals must be measured on both the predictor and the criterion. The indicator of validity is the degree of relationship between the predictor and criterion, and it is most often represented by the correlation coefficient. In this context a number of special issues surrounding criterion-related validity were discussed. Chief among them were the effects of unreliability and/or restriction of range on either the predictor or the criterion. There are "corrections" for the effects of unreliability and/or restriction of range that may be useful in some instances, but they must be used with extreme caution.

A major concern regarding criterion-related validity has to do with alternative ways to interpret or portray the usefulness of a particular magnitude of validity. The correlation coefficient is limited in this regard. In this chapter we discussed the rudiments of a *decisionmaking* and a *utility* approach to interpreting validity. The chief lesson to be learned is that the usefulness of a particular *validity coefficient* may vary considerably depending on a number of other characteristics of the situation (for example, base rate, selection ratio, values attached to different types of errors).

The second major theme in this chapter has to do with the validity of our understanding of what a particular set of operations measure. The focus here is not on the criterion, but on the meaning of the scores or the psychological variable of interest. *Content validity* and *construct validity* were discussed in this regard, and a variety of methods that can be used to establish such validities were considered. The touchstone of content validity is the judgment of experts regarding the degree to which the content of a measure constitutes a representative sample of the population of knowledges, skills, behaviors, or a variety of methods, some of which involve making predictions about how scores on the measure will be related to other observable events. The degree of construct validity is reflected by the degree of convergence of positive results from a series of relatively "demanding" empirical tests.

Finally we tried in this chapter to use what we have learned about the validity concept to illuminate some of the issues surrounding the problem of providing valid and nondiscriminatory prediction in the contexts

of educational and employment selection. These issues are not easily resolved, and they incorporate both technical questions and value judgments. If only we can keep these two considerations distinct and not shy away from either, we will have achieved much.

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